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Solution

English

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BPSC TEST-29/13 (SOLUTION)

Q 1. Ans. A

Explanation:

- **Statement 1 is correct:** Assaka (or Asmaka) was the southernmost of the sixteen Mahajanapadas and the only one situated entirely south of the Vindhya Range. It was located on the banks of the Godavari River, in the region that corresponds to modern-day Maharashtra and Telangana, with its capital at Potali (or Podana).
- **Statement 2 is incorrect:** Vajji (or Vrijji) was not a monarchy; it was a prominent confederacy of eight republican clans (Gana-Sangha), the most famous of which were the Licchavis. Furthermore, its capital was **Vaishali**, not Champa. Champa was the capital of the Anga Mahajanapada.
- **Additional Facts:**
  - ✓ **Literary Sources:** The list of the sixteen great states (Solasa Mahajanapadas) that flourished in the 6th century BCE is most famously found in the Buddhist text, the Anguttara Nikaya, as well as in the Jain text, the Bhagavati Sutra.
  - ✓ **Two types of government:** The political landscape of this era was divided into two distinct systems:
    - **Monarchies (Rajyas):** Centralised states ruled by hereditary kings, such as Magadha, Kosala, Vatsa, and Avanti.
    - **Republics (Gana-Sanghas):** Oligarchies where decisions were made by a council or an assembly of clan heads. Vajji and Malla were the most prominent examples of this system.
  - ✓ **Rise of Magadha:** Among all the Mahajanapadas, Magadha eventually emerged as the most powerful, absorbing its neighbours and laying the foundation for India's first great empires (the Nanda and Mauryan empires). Its success was largely due to strategic geographical advantages, such as rich iron ore deposits for weapons and proximity to major trade routes.

Q 2. Ans. D

Explanation:

- **Statement 1 is correct:** Bimbisara (Haryanka dynasty) systematically used matrimonial alliances to secure his borders and expand his influence. His marriages included Kosaladevi (yielding the revenue of a Kashi village as dowry), Chellana (a Licchavi princess), and Khema (princess of Madra). He also built and fortified the capital city of Rajagriha (Girivraja).
- **Statement 2 is correct:** Shishunaga's most significant military achievement was the destruction of the Pradyota dynasty of Avanti, decisively ending a century-long rivalry. Following this consolidation of power, he temporarily shifted his capital from Vaishali.
- **Statement 3 is correct:** Mahapadma Nanda overthrew the last Shishunaga rulers. He assumed titles like Ekarat (sole sovereign) and Sarvakshatrantaka (destroyer of all Kshatriyas). His conquest of Kalinga is historically corroborated by the Hathigumpha inscription of King Kharavela (from a later era), which explicitly records a Nanda king excavating a canal in Kalinga and taking away the sacred Kalinga Jina idol as a war trophy.

Q 3. Ans. B

Explanation:

- **Pair 1 is correctly matched:** Ashoka sent his son Mahinda and daughter Sanghamitta to Tambapanni (ancient Sri Lanka) to propagate Buddhism. Their mission successfully converted the Sri Lankan king, Devanampiya Tissa.
- **Pair 2 is correctly matched:** According to the Sri Lankan chronicles (Mahavamsa and Dipavamsa), the monk Majjhantika (or Madhyantika) was dispatched to the regions of Kashmir and Gandhara.
- **Pair 3 is incorrectly matched:** Moggaliputta Tissa did not travel to Suvarnabhumi. He was the prominent monk who presided over the Third Buddhist Council at Pataliputra and was responsible for organising and dispatching all these missions. The actual missionaries sent to Suvarnabhumi (the "Golden Land," generally identified with parts of modern Myanmar and Thailand) were the monks **Sona and Uttara**.

Q 4. Ans. C

Explanation:

- **Barabar Caves → 1 (Earliest rock-cut caves with Mauryan polish):** Located in Jehanabad district, Bihar, and commissioned by Ashoka (Sudama Cave dedicated in 261 BCE) and later his grandson Dasharatha, the Barabar Caves are the earliest surviving rock-cut caves in India, renowned for their distinctive mirror-like "Mauryan polish" on granite surfaces — a technique unmatched even by later cave complexes like Ajanta and Ellora.

- **Mauryan Ringstones → 2 (Miniature stone discs linked to fertility worship):** Ringstones are small, finely carved circular stone discs from the Mauryan period, generally believed by art historians to be associated with fertility/mother-goddess worship, often featuring rings of floral or figural motifs around a central perforation.
- **Sarnath Capital → 3 (Four lions with decorated abacus):** The Lion Capital of Ashoka's pillar at Sarnath (now India's National Emblem) features four addorsed lions standing on an abacus decorated with a dharma chakra alongside relief carvings of a bull, horse, elephant, and lion — symbolising the four cardinal directions and Buddhist teachings.
- **Dhauli Elephant → 4 (Rock-cut elephant carved from a boulder):** At Dhauli, Odisha (near the site of the Kalinga War), a rock-cut elephant — one of the earliest surviving animal sculptures in Indian art — is carved emerging from a natural boulder, near Ashoka's rock edicts, symbolising the Buddha (the "Great Being").

Q 5. Ans. A

Explanation:

- **Statement 1 is correct:** The Gandhara School (flourishing in the northwest frontier) is renowned for creating some of the earliest anthropomorphic representations of the Buddha. Its defining features are strong Greco-Roman (Hellenistic) stylistic influences and the use of bluish-grey schist stone (and later stucco).
- **Statement 2 is incorrect:** The Mathura School predominantly used **spotted red sandstone** (not white limestone). Furthermore, features such as a moustache, wavy hair, and heavy, thick drapery (resembling a Roman toga) are characteristic of the **Gandhara School**. Early Mathura Buddhas were typically depicted with a shaved head or tight snail-shell curls, a smiling face without a moustache, and wearing light, almost transparent, tight-fitting garments.
- **Statement 3 is correct:** The Amaravati School, which flourished in the Krishna-Godavari valley under the patronage of the Satavahana (and later Ikshvaku) rulers, is celebrated for its dynamic narrative relief panels. These sculptures were distinctively carved from **white limestone** (Palnad marble) to adorn stupas.

Q 6. Ans. C

Explanation:

- **Pair 1 is correctly matched:** The Dashavatara Temple at Deogarh (Uttar Pradesh) is one of the earliest surviving Hindu stone temples. Built during the Gupta period, it marks a critical architectural evolution by featuring an early, rudimentary shikhara (tower) and employing a Panchayatana layout (a main central shrine surrounded by four smaller subsidiary shrines at the corners).
- **Pair 2 is correctly matched:** The Sultanganj Buddha is a monumental (over 2 meters tall) copper statue dating back to the Gupta era (approx. 5th-7th century CE). It is a masterclass in ancient Indian metallurgy, cast seamlessly using the cire perdue (lost-wax) technique.
- **Pair 3 is correctly matched:** Cave 1 at Ajanta (dating to the late 5th century under Vakataka patronage, a contemporary and ally of the Guptas) houses some of the most celebrated murals in Indian art history, most notably the exquisite, life-size frescoes of the Bodhisattvas Padmapani (holding a lotus) and Vajrapani (holding a thunderbolt).

Q 7. Ans. A

Explanation:

- **Maukharis (4):** Following the decline of the Guptas, the Maukharis established their capital at **Kanauj**, transforming it into the premier political and strategic centre of North India (a status it retained for centuries).
- **Maitrakas (3):** This dynasty ruled the Saurashtra region in Gujarat from their capital at **Vallabhi**. Under their patronage, Vallabhi grew into a prosperous port city and a renowned centre for Buddhist learning, rivalling Nalanda.
- **Pushyabhutis (1):** Also known as the Vardhana dynasty, their ancestral capital was **Thanesar** (in present-day Haryana). Harshavardhana later united the Thanesar and Kanauj kingdoms, shifting his seat of power entirely to Kanauj.
- **Vakatakas (2):** Ruling over the Deccan (specifically the Vidarbha region), the Vakataka empire was divided into two main branches: the Pravarapura-Nandivardhana branch and the **Vatsagulma** branch.

Q 8. Ans. C

Explanation:

- The **Rampurva Capital** (specifically the Rampurva Bull Capital) is renowned for its masterfully carved, life-size single bull standing on a bell-shaped lotus base. Discovered at Rampurva in Bihar, it stands out because most surviving Ashokan pillars (like those at Sarnath, Sanchi, and Lauriya Nandangarh) are crowned with single or multiple lions. The Rampurva Bull is celebrated for its naturalistic modelling and heavy proportions, representing a pinnacle of Mauryan sculptural art.

Q 9. Ans. C

Explanation:

- **Guru Arjan Dev, the fifth Sikh Guru, is credited with two foundational structural developments in the Sikh faith:**
  - ✓ **Compilation of the Adi Granth (1604):** He compiled the writings of the first four Gurus and his own, alongside the verses of prominent Hindu Bhakti saints (like Kabir, Namdev, and Ravidas) and Islamic Sufi mystics (like Baba Farid). This was a radical step in religious pluralism, creating a unified scripture.
  - ✓ **Formalising the Masand System:** While Guru Ram Das Ji initiated the concept to gather voluntary contributions for Amritsar's development, Guru Arjan Dev formalised the masand system to manage the growing Sikh community and fund large-scale projects like the construction of the Harmandir Sahib (Golden Temple) in Amritsar. He institutionalised the Dasvandh, a system where Sikhs contributed one-tenth of their income. The Masands were the authorised representatives who collected this tithe and spread the faith.
- **Additional Facts :**
  - ✓ **Guru Angad (2nd Guru):** He standardised and popularised the **Gurmukhi script**. This was highly strategic, and it broke the monopoly of the priestly class over Sanskrit texts and allowed the common people to read the Guru's hymns in their spoken language. He also heavily expanded the Langar (community kitchen) system to enforce caste equality.
  - ✓ **Guru Ram Das (4th Guru):** He laid the foundation for the city of **Amritsar** (initially called Ramdaspur) on land donated by the Mughal Emperor Akbar, and began the excavation of the holy Sarovar (pool of nectar). Culturally, he composed the Lavan, the four-stanza hymn that forms the core of the Anand Karaj (the standard Sikh marriage ceremony).
  - ✓ **Guru Hargobind (6th Guru):** Following the execution of Guru Arjan Dev by the Mughal Emperor Jahangir, he transformed the Sikh community into a martial brotherhood. He introduced the doctrine of **Miri-Piri** (wearing two swords representing temporal power and spiritual authority) and constructed the **Akal Takht** (Throne of the Timeless One) directly opposite the Harmandir Sahib to handle political and military affairs.

Q 10. Ans. C

Explanation:

- The principal objective of the **Mahzar** (often referred to as the Infallibility Decree), issued in September 1579, was to vest the final authority on disputed questions of Islamic law in the Emperor himself.
- Drafted by **Sheikh Mubarak** (the father of Abul Fazl and Faizi), the document declared Akbar as the **Imam-i-Adil** (Just Ruler). It was a strategic political move to break the monopoly and excessive influence of the orthodox Islamic clerics (the Ulema), who were constantly bickering among themselves. The Mahzar stipulated that if there was a conflict of opinion among the religious scholars, Akbar had the supreme right to choose any interpretation that he believed was for the benefit of the state and the people.
- **Additional Facts:**
  - ✓ **Ibadat Khana (1575):** The "House of Worship" at Fatehpur Sikri was initially built for Sunni theologians to debate. The fierce, ugly theological disputes there disgusted Akbar and directly prompted him to issue the Mahzar. After 1578, he opened the Ibadat Khana to scholars of all religions, Hindus, Jains, Christians (Jesuits), and Zoroastrians.
- **Din-i-Ilahi (1582):** Din-i-Ilahi (Tauhid-i-Ilahi) was not a formal religion with a priesthood or holy book, but rather an ethical code and spiritual brotherhood for his closest courtiers (like Birbal), focusing on virtues like piety, prudence, and abstinence.
- **Abolition of Discriminatory Taxes:** Akbar abolished the pilgrim tax on Hindus in 1563 and the Jizya (tax on non-Muslims) in 1564, well before the Mahzar was issued. These early steps signalled his intent to create a more secular and inclusive administrative state.

Q 11. Ans. C

Explanation:

The first famous sedition trial in modern Indian history was initiated in 1891 against Jogendra Chunder

Bose, the editor of the vernacular Bengali weekly newspaper Bangobasi. Charged under Section 124A of the Indian Penal Code—which had been added to the code in 1870 by Sir James Fitzjames Stephen to curb rising anti-colonial dissent—Bose was targeted for publishing articles that fiercely criticized the British administration's Age of Consent Act of 1891. During the landmark trial, Queen-Empress v. Jogendra Chunder Bose, Chief Justice Sir William Comer Petheram created a vital legal precedent by defining "disaffection" as an active feeling of enmity and disloyalty toward the government, distinct from mere political "disapprobation" or policy criticism. Although the trial ultimately ended without a unanimous jury verdict and the charges were dropped after an apology, it marked the inaugural deployment of the colonial sedition law against the Indian nationalist press.

Q 12. Ans. C

Explanation:

- The **London Indian Society**, founded in 1865 by Dadabhai Naoroji and W.C. Bonnerjee, was superseded just a year later by the **East India Association** in 1866.
- Dadabhai Naoroji spearheaded the creation of the East India Association to establish a broader, more impactful platform in London. Its primary goal was to discuss Indian issues and actively lobby British public opinion and Members of Parliament to advocate for Indian welfare. It became one of the most prominent pre-Congress political associations operating outside of India.
- **Additional Facts:**
  - ✓ **British Indian Association (1851):** Formed in Calcutta by merging the Landholders' Society and the Bengal British India Society. Radhakanta Deb was its first president. It primarily represented the interests of the orthodox Hindu elite and wealthy landlords, petitioning the British for Indian representation in the legislature.
  - ✓ **Indian Association (1876):** Founded by Surendranath Banerjee and Ananda Mohan Bose in Calcutta. Often seen as the most important pre-Congress nationalist organisation, its goal was to unify Indians on a common political program and create strong public opinion (notably leading the agitation against the lowering of the age limit for the ICS examination). It eventually merged with the Indian National Congress.
  - ✓ **Poona Sarvajanik Sabha (1870):** Established by Mahadev Govind Ranade, Ganesh Vasudeo Joshi, and others. It functioned as a mediating body between the government and the people, advocating for the legal rights of peasants.

Q 13. Ans. A

Explanation:

- **Statement 1 is correct:** In 1883, Kadambini Ganguly and Chandramukhi Basu became the first female graduates in the history of India (and the entire British Empire) when they received their Bachelor of Arts degrees from the University of Calcutta (Bethune College).
- **Statement 2 is correct:** She holds the distinction of being the first woman to address an open session of the Indian National Congress (INC).
- **Statement 3 is incorrect:** She did not address the INC at its first session in 1885. She addressed the Congress during its **sixth session in 1890**, held in Calcutta. (She was first a delegate to the INC in 1889, alongside five other women).
- **Additional Facts:**
  - ✓ **Medical Pioneer:** Beyond her BA degree, she was one of the first female Indian physicians trained in Western medicine. After sustained petitioning (spearheaded by her husband), she gained admission to the Calcutta Medical College in 1884. She later travelled to the UK to acquire further advanced medical diplomas (LRCP, LRCS, LFPS).
  - ✓ **Social Reform Connections:** She was married to Dwarkanath Ganguly, a prominent Brahmo Samaj leader and fierce advocate for women's emancipation. It was Dwarkanath who led the agitation to force Calcutta University to open its doors to female students.

Q 14. Ans. D

Explanation:

- **Statement 1 is correct:** The Moderates played a foundational role in popularising modern political ideas in India. They constantly championed civil liberties (like freedom of speech and the press), democratic ideas, and the gradual introduction of representative institutions.
- **Statement 2 is correct:** Perhaps their greatest intellectual achievement was the "Economic Critique of Colonialism." They systematically dismantled the myth of British benevolence, proving through data that British policies were actively deindustrialising India and draining its wealth.

- **Statement 3 is correct:** The early political pressure exerted by the INC (passing resolutions, sending petitions, and lobbying in London) directly forced the British government to pass the **Indian Councils Act of 1892**. This act expanded the size of both the central and provincial legislative councils, a key initial demand of the Congress.
- **Additional Facts:**
  - ✓ Dadabhai Naoroji articulated the "Drain Theory" in his landmark book Poverty and Un-British Rule in India (1901). Romesh Chunder (R.C.) Dutt reinforced this with his definitive work, The Economic History of India (1901–1903), blaming British land revenue policies for devastating Indian agriculture.
  - ✓ **Limitations of the 1892 Act:** While securing the expansion of the councils was a victory, the Moderates were highly dissatisfied with the Act. It did not grant the right to vote on the budget, nor did it introduce direct elections (the term "election" was carefully avoided in the Act, using "nomination" based on recommendations instead). Their subsequent slogan became "No taxation without representation."
  - ✓ **Defence of Civil Rights:** When Bal Gangadhar Tilak (an Extremist leader) was arrested in 1897 for sedition, Moderate leaders, despite their sharp ideological differences with him, fiercely defended him and condemned the government's assault on the freedom of the press.

Q 15. Ans. C

Explanation:

- **Anusilan Samiti (1):** This was a prominent secret revolutionary organisation established in **Bengal** in 1902 by Pramathanath Mitra (P. Mitra), along with Jatindranath Banerjee and Barindra Kumar Ghosh, to overthrow British rule.
- **Abhinava Bharat (2):** Founded in 1904 in Maharashtra by the **Savarkar brothers** (Vinayak Damodar Savarkar and Ganesh Damodar Savarkar). It was a secret society of revolutionaries.
- **Bharathmatha Association (3):** Founded by **Nilakanta Brahmachari** in 1904 in the Madras Presidency. It was a secret society aimed at instigating a revolt against the British, famously associated with Vanchi Iyer.
- **Ghadar Party (4):** Originally formed in Astoria, Oregon, in 1913 and headquartered at the Yugantar Ashram in San Francisco, the Ghadar Party was an international political movement organised by expatriate Indians to overthrow British colonial rule through armed revolution.
- **Additional Facts:**
  - ✓ **Evolution of Abhinava Bharat:** The organisation originally started as **Mitra Mela** in 1899 in Nasik before being renamed and restructured into the Abhinava Bharat Society in 1904, heavily inspired by Giuseppe Mazzini's "Young Italy" movement.
  - ✓ **Ashe Assassination:** The Bharathmatha Association's most famous act was carried out by **Vanchi Iyer,** who assassinated **Robert Ashe** (the District Collector of Tirunelveli) in 1911 at the Maniyachi railway station before turning the gun on himself.
  - ✓ The headquarters of the Ghadar Party in San Francisco was known as the **Yugantar Ashram**, named after the influential Bengali revolutionary nationalist weekly Yugantar.

Q 16. Ans. C

Explanation:

- **Statement 1 is correct:** The Barrah Dacoity of 1908 was a major political robbery organised by the Dacca branch of the Anushilan Samiti under the leadership of **Pulin Behari Das**. These political dacoities (Swadeshi dacoities) were conducted to raise funds for purchasing arms and sustaining the revolutionary movement.
- **Statement 2 is correct:** In April 1908, in Muzaffarpur (Bihar), revolutionaries **Prafulla Chaki and Khudiram Bose** threw a bomb at a carriage they believed was occupied by D.H. Kingsford, the unpopular former Chief Presidency Magistrate of Calcutta, who was notorious for handing out harsh, cruel punishments to nationalists. Unfortunately, the carriage was carrying two British women (the wife and daughter of barrister Pringle Kennedy), both of whom were killed.
- **Additional Facts:**
  - ✓ Prafulla Chaki shot himself dead at the Mokama railway station to evade capture. Khudiram Bose, who was only 18 years old, was arrested, tried, and hanged, making him one of the youngest and most celebrated martyrs of the Indian independence movement.
  - ✓ **Alipore Bomb Conspiracy Case (1908):** The Muzaffarpur bombing triggered massive police raids, notably at the Maniktala gardens

in Calcutta. This resulted in the arrest of several prominent Anushilan Samiti leaders, including **Aurobindo Ghosh and his brother Barindra Kumar Ghosh**. Aurobindo was famously defended by Chittaranjan Das (C.R. Das) and was acquitted, after which he retired from active politics to an ashram in Pondicherry. Barindra Ghosh was sentenced to life imprisonment in the Cellular Jail (Kalapani).

Q 17. Ans. D

Explanation:

- All the listed demands were central to the early political and economic programme of the Moderate phase (1885–1905) of the Indian National Congress.
  - ✓ **Reduction of land revenue:** The Moderates identified excessive land revenue as the primary cause of rural poverty and famines. Leaders like R.C. Dutt vigorously campaigned for a permanent settlement of land revenue to protect peasants from arbitrary hikes.
  - ✓ **Abolition of the salt tax and sugar duty:** Long before Gandhi's Dandi March, the Moderates consistently condemned the salt tax as an unjust and regressive burden on the absolute poorest sections of Indian society.
  - ✓ **Freedom of speech and expression:** The defence of civil rights was a cornerstone of Moderate politics. They constantly petitioned against press censorship and arbitrary arrests, viewing freedom of speech as a fundamental right of modern citizenship.
  - ✓ **Reduction in military expenditure:** The Moderates harshly criticised the colonial government for spending nearly half of its revenues on the British Indian Army, which was frequently used to fight imperial wars abroad (like in Burma or Afghanistan) at the expense of Indian taxpayers.
- **Additional Facts:**
  - ✓ **The Welby Commission (1895):** The relentless campaigning by the Moderates regarding the financial drain and excessive military expenditure forced the British government to appoint the Royal Commission on the Administration of Expenditure of India (the Welby Commission). Dadabhai Naoroji was the first Indian to sit on a Royal Commission and present extensive evidence before it.
  - ✓ **Indianisation of Civil Services:** Alongside military cuts, a major economic demand was holding simultaneous ICS examinations in India and England. They argued that the high salaries and pensions paid to British officials (who took their wealth back to England) constituted a massive drain of wealth, which could be mitigated by employing Indians.

Q 18. Ans. B

Explanation:

- The Defence of India Act, 1915, was specifically enacted to crush the Ghadar movement and was first invoked during the **First Lahore Conspiracy Trial** (1915).
- Following the outbreak of World War I, the Ghadarites (led by figures like Rash Behari Bose and Sachindra Nath Sanyal) attempted to trigger a pan-Indian armed mutiny in the British Indian Army in February 1915. The British intelligence infiltrated the movement and pre-empted the rebellion. To swiftly deal with the captured revolutionaries without the slow process of standard common law, the Governor-General rushed the Defence of India Act through the legislative council in March 1915. It allowed for preventive detention, internment without trial, and the creation of special tribunals whose verdicts could not be appealed.
- **Additional Facts:**
  - ✓ **Special Tribunals:** Under the Act, the First Lahore Conspiracy Trial was conducted by a special tribunal of three commissioners. Out of the hundreds arrested, 24 Ghadarites were sentenced to death, and over 100 were sentenced to life imprisonment in the Cellular Jail (Kalapani).
  - ✓ **Connection to the Rowlatt Act:** The Defence of India Act was a temporary wartime measure meant to expire six months after the end of World War I. However, the British colonial government found its draconian powers so useful for crushing nationalist dissent that it decided to make its provisions permanent in peacetime. This peacetime extension was the infamous **Rowlatt Act (1919)**.

Q 19. Ans. A

Explanation:

- **Statement 1 is correct:** The proposed Ilbert Bill sparked massive outrage among the British and Anglo-Indian community in India. In response, they organised fiercely, raised funds, and formed the **European and Anglo-Indian Defence Association** to block the legislation.

- **Statement 2 is incorrect:** The British press in England did not support the bill. Heavily influenced by the lobbying of the Defence Association and reflecting the racial prejudices of the era, the English press widely condemned the bill and mocked Lord Ripon's administration for even proposing it.
- **Statement 3 is correct:** Faced with a near-mutiny from the white community and a lack of support from London, Lord Ripon was forced into a humiliating compromise. The modified bill stipulated that while Indian judges could try Europeans, the European defendant had the right to demand a jury where at least half of the members were European or American.
- **Additional Facts:**
  - ✓ Drafted by Sir Courtenay Ilbert, the Law Member of the Viceroy's Executive Council, the bill aimed to remove judicial disqualifications based on race. At the time, British subjects outside major presidency towns could only be tried by British judges. The bill sought to allow senior Indian magistrates and sessions judges to preside over these cases.
  - ✓ **White Mutiny:** The European reaction was extraordinarily vitriolic and racist. Planters and merchants argued that Indian judges could not be trusted to understand the "complexities" of English life or treat English women with respect. There were even secret plots among European elites in Calcutta to kidnap Lord Ripon and put him on a ship back to England.
  - ✓ The Ilbert Bill controversy was a watershed moment for the Indian intelligentsia. Seeing how easily a small, organised group of Europeans could force the powerful Viceroy to back down taught Indian leaders the vital importance of coordinated, national-level political agitation. This realisation directly accelerated the founding of the **Indian National Congress in 1885**.

Q 20. Ans. D

Explanation:

- **The Non-Cooperation Movement, officially launched on August 1, 1920, was India's first nationwide mass movement under Mahatma Gandhi's leadership. It was driven by three primary grievances:**
  - ✓ **Resentment with the Rowlatt Act (3):** The passing of the Anarchical and Revolutionary Crimes Act of 1919 (Rowlatt Act), which allowed the government to imprison political activists without trial for up to two years, deeply betrayed Indian leaders who had supported the British during WWI in hopes of constitutional concessions.
  - ✓ **"Punjab Wrongs" of 1919 (1):** The protests against the Rowlatt Act culminated in the horrific Jallianwala Bagh massacre on April 13, 1919. The subsequent imposition of martial law in Punjab, where Indians were subjected to humiliating punishments (like the "crawling order"), and the Hunter Commission's failure to adequately punish General Dyer, convinced Gandhi that the British government had lost its moral right to rule.
  - ✓ **The Khilafat Wrong (2):** Following WWI, the British imposed the harsh Treaty of Sèvres on the defeated Ottoman Empire, dismembering it and threatening the spiritual authority of the Caliph (Khalifa). This deeply angered Indian Muslims. Gandhi seized this as a once-in-a-century opportunity to cement Hindu-Muslim unity by merging the Khilafat and Non-Cooperation movements against the British.
- **Additional Facts:**
  - ✓ **Special Calcutta Session (Sept 1920):** This is where Gandhi moved the landmark Non-Cooperation resolution. It faced initial resistance from veteran leaders like C.R. Das and Annie Besant, who opposed the boycott of legislative councils, but the resolution passed.
  - ✓ **Nagpur Session (Dec 1920):** The movement was formally ratified here. More importantly, the Congress underwent a radical organisational restructuring: Provincial Congress Committees were reorganised on a linguistic basis, and the membership fee was reduced to just four annas per year, successfully transforming the INC from an elite debating club into a true mass organisation.

Q 21. Ans. A

Explanation:

- **Statement 1 is correct:** On February 4, 1922, a large group of volunteers participating in the Non-Cooperation Movement gathered in Chauri Chaura (in the Gorakhpur district of the United Provinces) to protest against high food prices and liquor sales. The local police beat some of the volunteers. When a larger crowd marched to the police station to protest this, the police fired warning shots into the air and eventually into the crowd, killing a few protesters. This police firing is what triggered the violent retaliation.

- **Statement 2 is incorrect:** Enraged by the police firing, the mob chased the heavily outnumbered policemen back into the police station (thana) and set the building on fire. When policemen tried to escape the flames, they were hacked to death or thrown back into the fire. A total of 22 policemen and 3 civilians died.
- **Additional Facts:**
  - ✓ **Bardoli Resolution:** Profoundly shocked by the violence and convinced that the Indian masses had not yet understood his strict principles of Ahimsa (non-violence), Mahatma Gandhi unilaterally called off the entire Non-Cooperation Movement on February 12, 1922, through the Bardoli Resolution.
  - ✓ Gandhi's sudden withdrawal of the movement while it was at its peak infuriated many top Congress leaders, including C.R. Das, Motilal Nehru, Subhas Chandra Bose, and Jawaharlal Nehru. This ideological split directly led to C.R. Das and Motilal Nehru breaking away to form the **Swaraj Party** in 1923, which advocated for entering the legislative councils to disrupt the government from within.
  - ✓ **Gandhi's Arrest:** Shortly after calling off the movement, Gandhi was arrested by the British in March 1922, charged with sedition, and sentenced to six years in prison (though he was released early in 1924 on medical grounds).

Q 22. Ans. C

Explanation:

- **Statement 1 is correct:** During the municipal elections of 1923–24, Congressmen achieved massive electoral success and captured numerous municipalities and local boards across the country. Prominent leaders took the helm of major cities: C.R. Das became the Mayor of Calcutta (with Subhas Chandra Bose as his CEO), Vithalbhai Patel headed the Bombay Corporation, and Jawaharlal Nehru became the Chairman of the Allahabad Municipal Board.
- **Statement 2 is correct:** The Congress at this time was divided into "Swarajists" and "No-Changers." While the No-Changers vehemently opposed entering the central and provincial legislative councils (viewing it as a political compromise), they willingly participated in local and municipal elections. They believed that local bodies, which handled civic duties like education, health, and sanitation, could be practically utilised to implement Gandhi's "constructive programme" (promoting khadi, national education, and untouchability eradication). For example, staunch No-Changers like Rajendra Prasad and Vallabhbhai Patel headed the Patna and Ahmedabad municipalities, respectively.
- **Additional Facts:**
  - ✓ **Gaya Split (1922):** Following the abrupt halt of the Non-Cooperation Movement, the Congress split at the Gaya session. C.R. Das and Motilal Nehru (Pro-Changers/Swarajists) wanted to end the boycott of legislative councils to disrupt the British government from within ("end or mend"). The orthodox Gandhians (No-Changers), led by C. Rajagopalachari, Vallabhbhai Patel, and Rajendra Prasad, wanted to stick to the boycott and focus purely on grassroots constructive work.
  - ✓ **Belgaum Compromise (1924):** Desperate to avoid another disastrous split like the one at Surat in 1907, both factions compromised. At the 1924 Belgaum session (notably, the only Congress session ever presided over by Mahatma Gandhi), it was agreed that the Swarajists would work in the councils as an integral part of the Congress, while the No-Changers would manage the rural constructive work.
  - ✓ **Swarajist Legislative Victories:** The Swarajists achieved significant disruptions in the Central Legislative Assembly by forming a coalition to outvote the government on various budgets and draconian bills. Their crowning parliamentary achievement was the election of Swarajist leader **Vithalbhai Patel** as the first Indian President (Speaker) of the Central Legislative Assembly in 1925.

Q 23. Ans. C

Explanation:

- **Statement 1 is correct:** Following the defeat of their council entry resolution at the Gaya session of the Congress in December 1922, C.R. Das (who resigned as Congress President) and Motilal Nehru officially formed the Congress-Khilafat Swarajya Party (commonly known as the Swaraj Party) on **January 1, 1923**. C.R. Das served as the President, and Motilal Nehru as the Secretary.

- **Statement 2 is correct:** The fundamental ideology of the Swarajists was to carry the non-cooperation movement into the legislative councils. They argued for council entry not to cooperate with the colonial administration, but to expose the sham of the 1919 reforms by offering "uniform, continuous and consistent obstruction." Their goal was to wreck the government from within or force it into political deadlock.
- **Additional Facts:**
  - ✓ **Electoral Success:** In the November 1923 elections, the Swarajists performed exceptionally well. They won 42 out of 101 elected seats in the Central Legislative Assembly, emerging as the single largest block. They also secured a clear majority in the Central Provinces and emerged as the largest party in Bengal.
  - ✓ **The Split (Responsivists vs. Non-Responsivists):** The unity of the Swaraj Party did not last. By 1924–1926, a faction known as the "Responsivists" (led by figures like Madan Mohan Malaviya, N.C. Kelkar, and B.S. Moonje) argued for cooperating with the government to protect Hindu interests and accepting office. Motilal Nehru fiercely opposed this, leading to a bitter split.
  - ✓ **Decline:** The sudden death of the charismatic C.R. Das in June 1925 dealt a fatal blow to the party. Combined with internal communal divisions, the Responsivist split, and the eventual call by the Congress in 1929 to boycott the councils and demand Purna Swaraj (complete independence), the Swaraj Party gradually withered away and formally merged back into the mainstream Congress.

Q 24. Ans. B

Explanation:

- The Belgaum session of 1924 is historically significant because it remains the **only session of the Indian National Congress ever presided over by Mahatma Gandhi**.
- This session was crucial for party unity. Gandhi used his presidency to broker the "Belgaum Compromise" between the Swarajists (who wanted to enter the legislative councils) and the No-Changers (who wanted to stick to rural constructive work and boycott the councils). By officially recognising the Swaraj Party's council work as a legitimate part of the broader Congress political programme, Gandhi successfully averted another disastrous split in the party (like the 1907 Surat split).
- **Additional Facts:**
  - ✓ **Bal Gangadhar Tilak:** Despite being one of the most prominent mass leaders of the independence movement, Bal Gangadhar Tilak **never** presided over a session of the Indian National Congress.
  - ✓ **Poorna Swaraj:** The historic Poorna Swaraj (Complete Independence) resolution was adopted at the **Lahore Session in 1929**, which was presided over by Jawaharlal Nehru.
  - ✓ **Formation of the Swaraj Party:** The Swaraj Party was formed by C.R. Das and Motilal Nehru on **January 1, 1923**, following the defeat of their council entry resolution at the Gaya Session in December 1922 (presided over by C.R. Das).

Q 25. Ans. D

Explanation:

- **Statement 1 is correct:** The **Paris Indian Society** was founded in 1905 as a branch of the Indian Home Rule Society. It was established by Madam Bhikaji Cama, S.R. Rana, and Munchershah Burjorji Godrej under the patronage of Shyamji Krishna Varma. It became a major hub for Indian revolutionaries in Europe.
- **Statement 2 is correct:** On August 22, 1907, Madam Bhikaji Cama made history by unfurling the first version of the Indian national flag (the "Flag of Indian Independence") at the International Socialist Conference in **Stuttgart, Germany**.
- **Additional Facts:**
  - ✓ **"Mother of the Indian Revolution":** Madam Cama was a fierce advocate for Indian freedom and women's rights abroad. Because of her tireless international lobbying and support for armed rebellion, she earned this title. She earlier served as the private secretary to Dadabhai Naoroji during his time in London.
  - ✓ **Design of the 1907 Flag:** The flag she unfurled in Stuttgart (co-designed with Shyamji Krishna Varma) had three horizontal stripes: green at the top, saffron/yellow in the middle, and red at the bottom. The green stripe had eight blooming lotuses (representing the eight provinces of British India); the middle stripe had "Vande Mataram" written in Devanagari script; and the bottom stripe featured a sun on the left and a crescent moon and star on the right (symbolising Hindu-Muslim unity).
  - ✓ **Revolutionary Publications:** She financed and distributed revolutionary literature that was smuggled into India. She famously

published the journal **Bande Mataram** from Paris, and another called **Madan's Talwar** (named in honour of Madan Lal Dhingra, who assassinated William Hutt Curzon Wyllie).

Q 26. Ans. D

Explanation:

- **O. Hume:** Allan Octavian Hume was a retired British civil servant and the principal founder of the Indian National Congress in 1885. He served as its General Secretary for over two decades, but he **never** presided over any session of the Congress.
- **Henry Cotton:** Henry Cotton was a long-serving British civil servant in India who was highly sympathetic to the Indian nationalist cause. He presided over the INC during the **1904 Bombay session**.
- **Annie Besant:** A British theosophist and fierce advocate for Indian self-rule, she capitalised on the momentum of her Home Rule League to become the **first woman** to preside over the INC at the **1917 Calcutta session**.
- **Additional Facts:**
  - ✓ **George Yule (1888):** A Scottish merchant who became the very first foreign (and non-Indian) President of the INC at the Allahabad session. His appointment was a strategic move by the early Congress to show that their movement was secular, broad-based, and not anti-British, but anti-colonial policy.
  - ✓ **William Wedderburn (1889 & 1910):** A close associate of A.O. Hume and another retired ICS officer. He holds the distinction of being one of the few foreigners to preside over the Congress twice. He also championed the Indian cause in the British Parliament and later wrote Hume's biography.
  - ✓ **Alfred Webb (1894):** An Irish parliamentarian who presided over the Madras session. His presidency was significant because it highlighted the strong early solidarity and shared tactics between the Irish Home Rule movement and the Indian nationalist struggle.
  - ✓ **Nellie Sengupta (1933):** Born in Cambridge, England (as Edith Ellen Gray), she married Bengali revolutionary Jatindra Mohan Sengupta. When the British government arrested the elected Congress President (Madan Mohan Malaviya) before the 1933 Calcutta session, Nellie bravely stepped in to preside over it, becoming the third woman to do so.

Q 27. Ans. A

Explanation:

- **Kakori Conspiracy (Robbery) — 9 August 1925:** Members of the Hindustan Republican Association (HRA), including Ram Prasad Bismil and Ashfaqulla Khan, looted a train carrying government treasury money near Kakori (Uttar Pradesh) to fund revolutionary activities.
- **Reorganisation of HRA into HSRA — 8–9 September 1928:** Following the executions of Kakori-case leaders (Bismil, Ashfaqulla Khan, Roshan Singh, and Rajendra Lahiri, hanged in December 1927), surviving revolutionaries met at Feroz Shah Kotla, Delhi, and reconstituted the organisation as the Hindustan Socialist Republican Association (HSRA), reflecting Bhagat Singh's growing socialist/Marxist influence.
- **Assassination of Saunders — 17 December 1928:** In retaliation for the fatal lathi-charge on Lala Lajpat Rai during the anti-Simon Commission protest (30 October 1928), Bhagat Singh and Rajguru shot Assistant Superintendent of Police J.P. Saunders in Lahore (having mistaken him for Superintendent James Scott, their actual intended target), with Chandrashekhar Azad providing covering fire.
- **Bombing of the Central Legislative Assembly — 8 April 1929:** Bhagat Singh and Batukeshwar Dutt threw non-lethal bombs into the Assembly chamber in Delhi to protest repressive bills, deliberately courting arrest to use the ensuing trial as a public platform for HSRA's ideas — leading into the Lahore Conspiracy Case.

Q 28. Ans. B

Explanation:

- The statement "India is one vast prison-house. I repudiate this law and regard it as my sacred duty to break the mournful monotony of compulsory peace" was made by Mahatma Gandhi, addressed to Viceroy Lord Irwin in his famous letter dated 2 March 1930 — written just before the start of the Salt Satyagraha (Dandi March).
- In this letter, addressed formally as "Dear Friend," Gandhi laid out his grievances against British rule under the heading "Why do I regard British rule as a curse?" and gave Irwin advance notice that he and his companions would break the Salt Law starting 11 March 1930, unless the Viceroy addressed his eleven demands (covering issues like the salt tax, land revenue, military expenditure, and protective tariffs).

Q 29. Ans. B

Explanation:

- **Enactment of the Trade Union Act (1926):** This was a foundational piece of labour legislation in India. It provided legal recognition to registered trade unions and granted them immunity from certain civil and criminal liabilities for carrying out legitimate union activities.
- **Bombay Textile Mills Strike (1928):** A massive, historic six-month strike broke out in the Bombay textile mills. It was spearheaded by the communist-led **Girni Kamgar Union**, whose membership exploded during this period as they effectively paralysed the British-owned mills.
- **Enactment of the Trade Disputes Act (1929):** Alarmed by the success of the 1928 strikes, the colonial government passed this legislation in April 1929 to severely curb union power. The act made sympathetic strikes illegal and required mandatory advance notice for any strike in public utility services. (Note: It was against this Act and the Public Safety Bill that Bhagat Singh and Batukeshwar Dutt threw a bomb in the Central Legislative Assembly).
- **Meerut Conspiracy Case (1929–1933):** Fearing the rapid spread of communist ideology, the British cracked down heavily by arresting 31 prominent labour leaders and communists (including S.A. Dange, Muzaffar Ahmed, and three British communists) in March 1929. While the initial arrests occurred right around the time the Trade Disputes Bill was being debated, the ensuing trial became a highly publicised, multi-year saga that concluded in 1933, making it the culminating historical marker of this sequence.

Q 30. Ans. D

Explanation:

- During the Civil Disobedience Movement, M.N. Roy secretly returned to India from Europe in December 1930 under the alias "Dr. Mahmud" to avoid British intelligence. He travelled extensively through the United Provinces, particularly focusing on the Awadh region. There, he worked to popularise socialist ideas among the rural peasantry and was highly instrumental in inspiring and initiating a no-tax and no-rent campaign. He continued this underground work until he was tracked down and arrested by the British police in Bombay in July 1931.
- **Additional Facts:**
  - ✓ **International Communist Leader:** M.N. Roy was a towering figure in the international communist movement. He holds the unique distinction of founding the Mexican Communist Party (1919) and being a founding member of the Communist Party of India (in Tashkent, 1920). He even served on the praesidium of the Communist International (Comintern) and debated directly with Vladimir Lenin on colonial liberation strategies.
  - ✓ **Constituent Assembly Pioneer:** Long before it became the official demand of the Indian National Congress (in 1935), it was M.N. Roy who, in 1934, first put forward the concrete idea that a Constituent Assembly, formed by Indians, must be convened to draft the Constitution of an independent India.
  - ✓ **Radical Humanism:** In his later years, he grew deeply disillusioned with orthodox Marxism and the authoritarianism of the Soviet Union. He dissolved his Radical Democratic Party in 1948 and spent the rest of his life developing and promoting a new political philosophy known as **Radical Humanism** (or New Humanism). This philosophy placed individual freedom, rationalism, and morality at the absolute centre of political life.

Q 31. Ans. D

Explanation:

- Malaria is caused by the **protozoan parasite Plasmodium** and is transmitted by the bite of an infected female Anopheles mosquito.
- Typhoid is caused by the **bacterium Salmonella typhi** and is transmitted through contaminated food and water.
- Ringworm is a **fungal infection** caused by species of **Microsporum, Trichophyton, and Epidermophyton**.
- Pneumonia is **not caused by a protozoan**. It is caused by the **bacterium Streptococcus pneumoniae and Haemophilus influenzae.**

Q 32. Ans. B

Explanation:

- **Malaria** is caused by the **protozoan parasite Plasmodium**. One of the common species infecting humans is **Plasmodium vivax**. Hence, this pair is **correctly matched**.
- Typhoid is caused by **Salmonella typhi**. **Vibrio cholerae** causes **cholera**.
- Tuberculosis is caused by the bacterium **Mycobacterium tuberculosis**.

- **Plasmodium falciparum** causes the most severe form of **malaria**.
- Amoebiasis is caused by the protozoan **Entamoeba histolytica**.
- **Giardia lamblia** causes **giardiasis**, a different intestinal disease.

Q 33. Ans. A

Explanation:

- **Statement: Cyanobacteria (blue-green algae)** are **photosynthetic prokaryotic organisms** belonging to Kingdom Monera. They contain **chlorophyll a** and carry out oxygenic photosynthesis similar to green plants. However, since they are prokaryotes, they **lack membrane-bound organelles**.
- **Conclusion I: Cyanobacteria are prokaryotes**, and prokaryotic cells **do not possess membrane-bound organelles**, including chloroplasts. Their photosynthetic pigments are present on **infoldings of the plasma membrane (thylakoid-like membranes)** rather than inside chloroplasts. Therefore, although they perform photosynthesis, they **do not have chloroplasts**. Hence, the conclusion is incorrect.
- **Conclusion II:** Only **some prokaryotes**, such as **cyanobacteria** and certain **photosynthetic bacteria**, can perform photosynthesis. Many other prokaryotes, including **Escherichia coli, Salmonella, Lactobacillus, and Mycobacterium**, are **non-photosynthetic** and obtain nutrition through heterotrophic or chemosynthetic modes. Therefore, **photosynthesis is not a universal characteristic of prokaryotes**.

Q 34. Ans. B

Explanation:

- The power rating of an electrical appliance indicates the rate at which it consumes electrical energy when operated at its rated voltage. The relationship between power (P), voltage (V), and resistance (R) is given by the equation $P = V^2/R$. This equation is derived from Ohm's law and the expression for electric power.
- In this question, the electric bulb is rated **100 W, 250 V**, which means it consumes **100 watts of power** when connected to a **250 V** supply. To determine the resistance of the bulb, the formula is rearranged as $R = V^2/P$.
  - ✓ **Substituting the given values:**
  - ✓ $R = (250)^2/100$
  - ✓ $R = 62500/100$
  - ✓ $R = 625\ \Omega$
- Thus, the resistance of the bulb operating at its rated voltage is **625 Ω**.

Q 35. Ans. A

Explanation:

- When an object is placed at the **centre of curvature** (C = 2F) of a **concave mirror**, the image is formed **at the centre of curvature itself**. The image is **real, inverted, and of the same size as the object**.
  - ✓ **Given:**
  - ✓ Object distance, $u = -30$ **cm**
  - ✓ Focal length of concave mirror, $f = -15$ **cm**
  - ✓ Since the object distance is equal to **2f (30 cm = 2 × 15 cm)**, the object is placed at the **centre of curvature** (C).
  - ✓ Using the mirror formula,
  - ✓ $1/f = 1/v + 1/u$
  - ✓ Substituting the values,
  - ✓ $1/(-15) = 1/v + 1/(-30)$
  - ✓ $1/v = -1/15 + 1/30 = -1/30$
  - ✓ Therefore, $v = -30$ **cm**
- Magnification is given by,
  - ✓ $m = -v/u$
  - ✓ Substituting the values,
  - ✓ $m = -(-30)/(-30) = -1$
- The negative sign indicates that the image is **inverted**, while the magnitude **1** indicates that the image is **equal in size** to the object. Hence, the magnification produced by the concave mirror is **–1**.

Q 36. Ans. A

Explanation:

- **Statement: Cellulose** is a structural polysaccharide made up of long chains of **β-D-glucose** molecules joined by **β(1β4) glycosidic bonds**. It is the chief constituent of the **plant cell wall** and provides rigidity and mechanical strength.
- **Conclusion I:** This conclusion is incorrect because **starch**, not cellulose, is the **principal storage carbohydrate** in plants. Starch serves as an energy reserve and is stored in seeds, tubers, roots, and other plant organs. In contrast, **cellulose is a structural carbohydrate**, forming the cell wall and providing support to plant cells.
- **Conclusion II:** Humans **cannot digest cellulose** because the human digestive system lacks the enzyme **cellulase**, which is required to break **β(1β4) glycosidic bonds**. Consequently, cellulose acts as a **dietary fibre (roughage)**, aiding bowel movement rather than serving as a source of energy. Only certain microorganisms present in the digestive

systems of ruminants and termites produce cellulase and can digest cellulose efficiently. Hence, this conclusion is incorrect

Q 37. Ans. C

Explanation:

- Filariasis, also known as **elephantiasis**, is caused by the parasitic roundworms **Wuchereria bancrofti and Brugia malayi**. While Culex mosquitoes primarily transmit Wuchereria bancrofti, Brugia malayi is primarily transmitted by Mansonia mosquitoes (and some Aedes species). Since transmission occurs through a mosquito vector, filariasis is classified as a **vector-borne disease**.
- Dengue is caused by the **Dengue virus**, which belongs to the genus Flavivirus. The disease is transmitted by the bite of an infected **female Aedes aegypti mosquito**, which usually bites during the daytime. Symptoms include high fever, severe headache, joint and muscle pain, skin rashes, and in severe cases, dengue haemorrhagic fever. Because the virus is spread through a mosquito vector, dengue is also a **vector-borne disease**.
- Tuberculosis (TB) is caused by the bacterium **Mycobacterium tuberculosis**. Unlike malaria or dengue, TB is **not transmitted through insects or any other vector**. It spreads directly from an infected person to a healthy person through **airborne droplets** released when the infected individual coughs, sneezes, speaks, or spits. The bacteria primarily affect the lungs, although other organs may also be involved. Since no vector is involved in its transmission, tuberculosis is **not a vector-borne disease**.
- Malaria is caused by protozoan parasites of the genus **Plasmodium**, including **P. vivax and P. falciparum**. It is transmitted through the bite of an infected **female Anopheles mosquito**. After entering the human body, the parasite first multiplies in the liver and later infects red blood cells, causing periodic fever, chills, sweating, and anaemia. The dependence on the mosquito for transmission makes malaria a classic **vector-borne disease**.

Q 38. Ans. B

Explanation:

- The **myelin sheath** is a multilayered, lipid-rich insulating covering that surrounds the axon of many neurons. In the **peripheral nervous system (PNS)**, it is formed by **Schwann cells**, whereas in the **central nervous system (CNS)**, it is produced by **oligodendrocytes**.
- The primary function of the myelin sheath is to **increase the speed of nerve impulse conduction**. It acts as an electrical insulator and allows the nerve impulse to travel by **saltatory conduction**, in which the impulse jumps from one **Node of Ranvier** to the next rather than travelling continuously along the entire axon membrane. This greatly increases the speed and efficiency of impulse transmission while reducing energy expenditure.

Q 39. Ans. B

Explanation:

- **The pituitary gland** is known as the **"master gland"** because it secretes several hormones that regulate the growth, development, and secretory activities of many other endocrine glands in the body. It is a small, pea-sized gland located at the base of the brain and is connected to the **hypothalamus** by a stalk called the **infundibulum**. The pituitary itself is regulated by the hypothalamus.
  - ✓ **The anterior pituitary secretes important tropic hormones, including:**
  - ✓ **Thyroid-stimulating hormone (TSH)** – stimulates the thyroid gland to produce thyroid hormones.
  - ✓ **Adrenocorticotropic Hormone (ACTH)** – stimulates the adrenal cortex to secrete corticosteroids.
  - ✓ **Follicle Stimulating Hormone (FSH)** and **Luteinizing Hormone (LH)** – regulate the functions of the gonads (ovaries and testes).
  - ✓ **Growth Hormone (GH)** – promotes the growth of bones and body tissues.
  - ✓ **Prolactin (PRL)** – stimulates milk production after childbirth.
- Because it controls the activity of several other endocrine glands, the pituitary gland is traditionally called the **"master gland."**

Q 40. Ans. B

Explanation:

- **Statement: Gymnosperms** are seed-producing plants in which the **ovules are not enclosed by an ovary**. Consequently, after fertilisation, the seeds remain **naked** and are **not enclosed within fruits**.
- **Conclusion I: Gymnosperms are well-developed vascular plants**. They possess both **xylem and phloem** for the conduction of water, minerals, and food. In fact, gymnosperms are among the earliest vascular seed plants. Therefore, the statement that every gymnosperm lacks vascular tissue is false. Hence, the conclusion is incorrect

- **Conclusion II:** In **angiosperms**, the **ovary develops into the fruit after fertilisation**. Since **gymnosperms do not possess ovaries**, they cannot produce fruits, and their seeds remain exposed or "naked." Thus, the absence of an ovary directly explains why gymnosperms do not form fruits. Hence, this conclusion is correct.

Q 41. Ans. C

Explanation:

- The **Law of Reflection** states that the **angle of incidence is equal to the angle of reflection**. Both angles are measured from the **normal** drawn perpendicular to the reflecting surface at the point of incidence.
- **Given:**
  - ✓ Angle of incidence (**i**) = **38°**
  - ✓ Therefore, Angle of reflection (**r**) = **38°**
  - ✓ **The angle between the incident ray and the reflected ray is given by:**
  - ✓ **Angle between incident and reflected rays = i + r = 2i**
  - ✓ Substituting the given value, **Angle = 2 × 38° = 76°**
- Hence, the angle between the incident and reflected rays is **76°**.

Q 42. Ans. C

Explanation:

- **Passenger falls backwards when bus starts suddenly (a - 2):** When a stationary bus suddenly starts moving, the lower part of a passenger's body moves forward with the bus. However, the upper part of the body tends to remain in its initial state of rest due to the **inertia of rest**, causing the passenger to fall backwards.
- **Passenger falls forward when bus stops suddenly (b - 1):** When a moving bus abruptly stops, the lower part of the body comes to rest along with the bus. The upper part of the body, however, tries to maintain its state of motion due to the **inertia of motion**, resulting in a forward fall.
- **A gun recoils when a bullet is fired (c - 4):** When a gun fires, it exerts a forward force on the bullet (action). According to Newton's Third Law, the bullet exerts an equal and opposite force on the gun (reaction). This backward force is what causes the gun to recoil, illustrating **action and reaction**.
- **Stone released from circular motion flies tangentially (d - 3):** An object moving in a circle is constantly having its direction changed by a centripetal force. When that force is removed (like a string breaking), the object will continue in a straight, tangential line due to its tendency to maintain its current direction of motion, known as the **inertia of direction**.

Q 43. Ans. C

Explanation:

- A cyclic **process** is a thermodynamic process in which a system undergoes a sequence of changes and finally returns to its **initial state**. Since the initial and final states of the system are identical, all **state functions**, including **internal energy**, have the same value at the beginning and at the end of the cycle.
- Internal energy is a **state function**, which means its value depends only on the state of the system and not on the path followed to reach that state. Therefore, when a system completes one full thermodynamic cycle and returns to its original state, the **net change in internal energy is zero**.

Q 44. Ans. B

Explanation:

- A refrigerator is a heat engine operating in reverse. Its primary function is to **remove heat from a low-temperature region (inside the refrigerator)** and transfer it to a **high-temperature region (the surrounding room)**. Since this process is **against the natural direction of heat flow**, it cannot occur spontaneously and requires an external source of energy in the form of **mechanical or electrical work** supplied to the compressor.
- The **Second Law of Thermodynamics (Clausius Statement)** states that **heat cannot, by itself, flow from a colder body to a hotter body**. Therefore, a refrigerator uses external work to transfer heat from the cold compartment to the warmer surroundings. This is why electrical energy is required for the operation of refrigerators and air conditioners.

Q 45. Ans. B

Explanation:

- **Technetium (Tc, Atomic Number 43)** is the **first artificially produced element**. It was first produced in **1937** by **Carlo Perrier** and **Emilio Segrè** by bombarding molybdenum with deuterons in a cyclotron. Before its artificial production, no stable naturally occurring isotope of technetium was known, making it the first element to be synthesised artificially.

Q 46. Ans. C

Explanation:

- Members of the **grass family (Poaceae/Gramineae)** are monocotyledonous plants characterised by fibrous roots, narrow leaves with parallel venation, hollow stems (culms), and flowers arranged in spikelets.
  - ✓ **Sugarcane:** Belongs to the **grass family (Poaceae)**. It is a monocot plant cultivated mainly for sugar production.
  - ✓ **Wheat:** Belongs to the **grass family (Poaceae)**. It is one of the world's most important cereal crops and is a typical monocot.
  - ✓ **Bamboo:** Belongs to the **grass family (Poaceae)**. Although woody in appearance, bamboo is botanically a giant grass.
  - ✓ **Coriander:** Does **not** belong to the grass family. It belongs to the **Apiaceae (Umbelliferae)** family and is a dicotyledonous herb cultivated as a spice and leafy vegetable.

Q 47. Ans. A

Explanation:

- The stability of halogen oxides generally **increases with an increase in the oxidation state** of the halogen atom. This trend is due to a combination of thermodynamic and kinetic factors; higher oxidation states form more bonds with oxygen, which helps distribute the electron density more evenly and minimises destabilising lone-pair repulsions.
- **The oxidation state of chlorine in each of the given options:**
  - ✓ **Cl_2O (Dichlorine monoxide):** +1
  - ✓ **ClO_2 (Chlorine dioxide):** +4
    - ▪ **ClO_3 (Chlorine trioxide, generally exists as Cl_2O_6):** +6
  - ✓ **Cl_2O_7 (Dichlorine heptoxide):** +7
- Since chlorine has the lowest oxidation state (+1) in **Cl_2O**, it is the **least stable** oxide among the choices. Conversely, **Cl_2O_7** (with the highest oxidation state of +7) is the **most stable** oxide of chlorine.

Q 48. Ans. D

Explanation:

- An **alkali** is a **base that is soluble in water** and produces **hydroxide ions** (OH^-) in aqueous solution. Alkalis have a pH greater than 7 and can neutralise acids to form salt and water.
- Ethanol is an **alcohol**, not a base or an alkali. Although it contains an **–OH (hydroxyl) group**, this group is **covalently bonded** and does **not dissociate to release OH^- ions** in water. Therefore, ethanol does not exhibit alkaline properties and cannot neutralise acids like an alkali.

Q 49. Ans. A

Explanation:

- The **acidic strength of oxyacids of the same central atom increases with the oxidation state of the central atom and the number of oxygen atoms attached to it**. Additional oxygen atoms withdraw electron density through the **–I (inductive) effect**, weakening the O–H bond and making it easier to release a proton (H^+).
- **The acidic strength of the oxyacids of chlorine follows the order:**
  - ✓ $HClO < HClO_2 < HClO_3 < HClO_4$

Q 50. Ans. C

Explanation:

Ethers are a class of organic compounds in which an oxygen atom is bonded to two alkyl or aryl groups. One of the most common ethers is **diethyl ether (commonly called ether)**. It was historically used as a **general anaesthetic** because it can produce temporary unconsciousness and loss of sensation during surgical procedures. Although modern medicine has largely replaced ether with safer and more effective anaesthetics due to its high flammability and side effects, it remains an example of an anaesthetic in chemistry.

Q 51. Ans. D

Explanation:

- **Iron (Fe)** is an essential mineral element and is a vital constituent of **haemoglobin**, the oxygen-carrying pigment present in the red blood cells (RBCs) of humans and other vertebrates.
- Haemoglobin is a **conjugated protein** found in red blood cells. It contains a **heme group**, at the centre of which lies an **iron (Fe^2β) ion**. This iron atom reversibly binds with oxygen in the lungs to form **oxyhaemoglobin** and releases oxygen to body tissues. Thus, iron is an essential structural and functional constituent of haemoglobin.

Q 52. Ans. D

Explanation:

- In a hydroelectric power plant, water stored at a height possesses **gravitational potential energy**. As the water flows downward, this potential energy is converted into kinetic energy, which rotates turbines connected to generators. The generators then convert this mechanical energy into **electrical energy**. Thus, the overall conversion is the **potential energy of water to electrical energy**.

- A solar cell works on the **photovoltaic effect**, where sunlight falling on a semiconductor material produces an electric current. Therefore, a solar cell directly converts **light energy into electrical energy**.
- An electric motor converts **electrical energy into mechanical energy**. When an electric current passes through a coil placed in a magnetic field, a force acts on the coil, causing it to rotate. This rotational motion is a useful mechanical output.
- A dry cell produces electricity through **chemical reactions** occurring between the electrolyte and electrodes inside the cell. The stored **chemical energy** is converted into **electrical energy**, which powers electrical devices.

Q 53. Ans. B

Explanation:

- **Pair 1 is incorrect: frequency determines the pitch (shrillness)** of a sound, not its loudness. A sound with a **higher frequency** has a **higher pitch**, while a sound with a lower frequency has a lower pitch. Loudness depends on the amplitude of vibration.
- **Pair 2 is incorrect: amplitude determines the loudness** of a sound. Greater amplitude means the sound is louder, whereas a smaller amplitude produces a feeble sound. Amplitude has no effect on the pitch of the sound.
- **Pair 3 is correct:** The **time period** (T) is the time taken for one complete vibration, whereas **frequency** (f) is the number of vibrations per second. They are related by the expression: **T = 1/f.** Thus, the time period is the **reciprocal of frequency**, making this pair correctly matched.
- **Pair 4 is correct:** The **human ear** can normally hear sounds having frequencies between **20 Hz and 20,000 Hz (20 kHz)**. Sounds below **20 Hz** are called **infrasonic**, while those above **20,000 Hz** are called **ultrasonic**. Hence, this pair is correctly matched.

Q 54. Ans. A

Explanation:

- **Statement 1 is correct:** Covalent compounds are generally **poor conductors of electricity** because they do **not contain free-moving ions or free electrons.** Electrical conduction requires the presence of charged particles that can move freely. Since covalent compounds consist of neutral molecules, they cannot conduct electricity in the solid, molten, or aqueous state (except in a few special cases such as graphite or substances that ionise in water).
- **Statement 2 is incorrect:** Covalent compounds do not generally produce free-moving ions in solution. When most covalent compounds (like sugar or ethanol) dissolve in water, they separate into individual, electrically neutral molecules, not ions. It is ionic compounds, like sodium chloride, that dissociate into free-moving ions in solution.
    - ✓ Some polar covalent compounds, most notably acids like HCl, HNO3, and H2SO4, are composed of neutral molecules that actively ionise/dissociate into free-moving ions when dissolved in water, making their aqueous solutions excellent conductors of electricity.

Q 55. Ans. C

Explanation:

- **Liquefied Petroleum Gas** (LPG) is mainly a mixture of **propane** (C_3H_8) and **butane** (C_4H_{10}). These gases are **colourless and odourless**, making it difficult to detect a gas leak by smell alone.
- To ensure safety, a small quantity of **ethyl mercaptan (ethanethiol,** C_2H_5SH) is added to LPG. Ethyl mercaptan has a **strong, unpleasant smell**, which enables people to detect even a small leakage of LPG and take necessary precautions to prevent accidents such as fire or explosion.

Q 56. Ans. A

Explanation:

- **Statement 1 is correct:** Hard water does **not readily form lather with soap** because the calcium and magnesium ions react with soap to form an insoluble precipitate called **scum**. As a result, more soap is required before lather is produced. Therefore, this statement is correct.
- **Statement 2 is correct: Clark's method** is used to remove **temporary hardness** of water. In this method, **calcium hydroxide (slaked lime), $Ca(OH)_2$**, is added to hard water. It reacts with soluble bicarbonates of calcium and magnesium, converting them into insoluble precipitates that can be removed by filtration. Thus, this statement is correct.
- **Statement 3 is incorrect:** Permanent hardness is caused by **chlorides and sulphates** of calcium and magnesium. **Boiling removes only temporary**

hardness, which is due to bicarbonates. Permanent hardness **cannot be removed by boiling** and requires methods such as the **zeolite process, ion-exchange process**, or the addition of washing soda.

Q 57. Ans. C

Explanation:

- **A more reactive metal can displace a less reactive metal from its salt solution. This is known as a displacement reaction and is based on the reactivity series of metals. The relevant part of the reactivity series is:**
  - ✓ Zn > Fe > Pb > H > Cu > Hg > Ag
- **Since zinc (Zn) is more reactive than iron (Fe), it can displace iron from iron sulphate (FeSOβ) solution. The reaction is:**
  - ✓ $Zn + FeSO_4 \rightarrow ZnSO_4 + Fe$
- Thus, zinc replaces iron from its salt solution.

Q 58. Ans. B

Explanation:

Haven-1 is the **world's first planned commercial space station**, being developed by the American aerospace company **Vast Space**. The station is scheduled to launch in **early 2027** aboard a **SpaceX Falcon 9** rocket from **Cape Canaveral, Florida**. The mission will unfold in two phases: the **uncrewed deployment of the Haven-1 station**, followed by the **Vast-1 crewed mission**, which is expected to carry **up to four astronauts** on research expeditions lasting **10 to 30 days**. Designed as a **single-module space station**, Haven-1 measures **10.1 metres in length** and **4.4 metres in diameter**, and includes **private crew quarters, a scientific payload laboratory, and a domed observation window**. Its **life-support and communication systems** have been tested both in-house and in collaboration with **NASA**, while continuous **Starlink Wi-Fi connectivity** will provide uninterrupted communication throughout the mission.

Q 59. Ans. D

Explanation:

BlueBird Block-2 is a next-generation series of **commercial communication satellites** developed by **AST SpaceMobile** to provide **space-based 4G and 5G cellular broadband** directly to **standard smartphones**, eliminating the need for ground-based cellular towers or specialised satellite handsets. Each satellite weighs approximately **6,100 kg** and is equipped with a **223-square-metre phased-array antenna**, enabling nearly **ten times the bandwidth capacity** of the earlier **Block-1** satellites. The first prototype, **BlueBird 6**, was launched on **24 December 2025** aboard **ISRO's LVM3-M6** rocket from the **Satish Dhawan Space Centre, Sriharikota**. It became the **heaviest commercial communication satellite ever placed into Low Earth Orbit** (LEO) and the **heaviest payload launched from Indian soil**. Operating in **LEO**, the BlueBird Block-2 constellation aims to provide **near-100% geographic coverage**, allowing users to make **voice and video calls, send text messages, and access broadband internet** from virtually anywhere using ordinary mobile phones.

Q 60. Ans. A

Explanation:

- The **2025 Nobel Prize in Physics** was awarded to **John Clarke, Michel H. Devoret, and John M. Martinis** for their pioneering discoveries of **macroscopic quantum mechanical tunneling** and **energy quantisation in electric circuits**. Their work laid the scientific foundation for superconducting quantum circuits, which are central to the development of modern **quantum computing**.
- **Mary E. Brunkow, Fred Ramsdell, and Shimon Sakaguchi** received the **2025 Nobel Prize in Physiology or Medicine** for discovering how **regulatory T cells (Tregs)** maintain **peripheral immune tolerance**, preventing autoimmune diseases.
- **Susumu Kitagawa, Richard Robson, and Omar M. Yaghi** were awarded the **2025 Nobel Prize in Chemistry** for developing **metal–organic frameworks (MOFs)**, a class of porous materials with wide-ranging applications in gas storage, catalysis, and environmental technologies.
- **David Baker, Demis Hassabis, and John M. Jumper** were the recipients of the **2024 Nobel Prize in Chemistry**, not the 2025 Nobel Prize in Physics.

Q 61. Ans. C

Explanation:

- The **12th Bihar Provincial Conference** was held at **Bhagalpur** under the presidency of **Dr. Rajendra Prasad**.
- On **28 August 1920**, the Conference passed a resolution supporting **Mahatma Gandhi's Non-Cooperation Movement**.
- Bihar became one of the earliest provinces to officially endorse Gandhi's programme even before the Special Session of the Indian National Congress at Calcutta (September 1920).
- Leaders such as **Mazharul Haq, Shah Mohammad Zubair, Gorakh Prasad**, and **Ghulam Imam** supported the resolution.

Q 62. Ans. B

Explanation:

- **Kesaria Chaur** is a large natural wetland situated in **East Champaran district** of Bihar. It is an important freshwater wetland that supports rich biodiversity, especially **migratory and resident water birds**, fish, and aquatic vegetation. The wetland also plays a significant role in **groundwater recharge, flood moderation, and sustaining local livelihoods**. It lies near the historic **Kesaria Stupa**, one of the important Buddhist heritage sites in Bihar.
- The nearby **Kesaria Stupa**, associated with Lord Buddha, is also located in East Champaran.

Q 63. Ans. C

Explanation:

- **Bihar is home to exactly two officially inscribed UNESCO World Heritage Sites:**
  - ✓ **Mahabodhi Temple Complex at Bodh Gaya:** Inscribed in 2002, this complex marks the revered location where Gautama Buddha is said to have attained enlightenment.
  - ✓ **Archaeological Site of Nalanda Mahavihara:** Inscribed in 2016, this site contains the ruins of what was once a massive, internationally renowned ancient centre of learning and Buddhist monastery.
- The other two options are currently on UNESCO's **Tentative List**.
  - ✓ **Tomb of Sher Shah Suri at Sasaram:** This Indo-Islamic mausoleum, built in the middle of an artificial lake, has been waiting on the tentative list since 1998.
  - ✓ **Remains of Vikramshila Ancient University:** Established by the Pala king Dharmapala, the ruins of this ancient premier university were added to the tentative list in 2010 as part of the "Silk Road Sites in India."

Q 64. Ans. B

Explanation:

During the Non-Cooperation Movement, the visit of the **Prince of Wales** was opposed throughout India. When the Prince reached **Patna on 22 December 1921**, a complete **hartal (strike)** and boycott were observed. The boycott reflected Bihar's strong participation in Gandhi's Non-Cooperation Movement. Nationalist leaders encouraged people not to participate in official receptions.

Q 65. Ans. C

Explanation:

- **Statement 1 is correct:** The **Morhar** and **Dardha** are two of the **four major right-bank tributaries** of the **Punpun River**. The other important right-bank tributaries are the **Madar** and **Batane**. These tributaries originate from the **Chotanagpur hill region**, mainly in the **Palamu, Aurangabad, and Gaya** areas, and contribute a substantial volume of water during the southwest monsoon. Consequently, they play a **significant role in the flood behaviour** of the Punpun basin.
- **Statement 2 is correct:** The **Begi** and **Siroka** are **left-bank tributaries** of the Punpun River. Other left-bank tributaries include the **Khudwa** and **Panchane**. However, unlike the right-bank tributaries, these rivers are **small seasonal streams**. While they are recognised as the **major left-bank tributaries** of the Punpun in terms of drainage, **their contribution to flood discharge is insignificant** because of their comparatively small catchment areas and lower discharge.

Q 66. Ans. A

Explanation:

- According to **NITI Aayog's National Multidimensional Poverty Index 2023**, Bihar recorded the highest incidence of multidimensional poverty among Indian States. Approximately **33%** of Bihar's population remained multidimensionally poor.
  - ✓ **The MPI measures deprivation across:**
  - ✓ Health
  - ✓ Education
  - ✓ Standard of Living
- Although Bihar registered substantial improvement compared to the NFHS-4 period, it continued to have the highest poverty incidence.

Q 67. Ans. A

Explanation:

- The **Bihar Scientific Society** was established on **24 May 1868** at **Muzaffarpur**. It was founded by **Imdad Ali**.
- **The Society promoted:**
  - ✓ Scientific education
  - ✓ Translation of scientific literature into Indian languages
  - ✓ Dissemination of modern scientific knowledge among the public
- It played an important role in spreading scientific awareness during the colonial period in Bihar.

Q 68. Ans. A

Explanation:

- **Statement 1 is correct:** The **Mahananda River** originates from **Mohalidram Hill** in the **Himalayas** at **Chimali**, about **6.4 km northeast of Kurseong** in the **Darjeeling district of West Bengal**, at an elevation of approximately **2,060 metres**. It is a major **northern tributary of the Ganga** and flows through **India (West Bengal and Bihar), Bangladesh, and a small part of Nepal** before finally joining the **Padma (Ganga)** near **Godagrighat** in Bangladesh.
- **Statement 2 is correct:** After flowing about **20 km through the Darjeeling hills**, the Mahananda enters the plains near **Siliguri**. Just below Siliguri, the **Balason River** joins the **Mahananda on its right bank**. Other important right-bank tributaries include the **Old Balason**, **Chenge**, **Eastern Kankai**, and **Western Kankai**.
- **Statement 3 is incorrect:** Near **Bagdob in Bihar**, the Mahananda bifurcates into two channels: **Phulhar (Jhaua) branch** – carries about **75%** of the river's total discharge. **Barsoi branch** – carries the remaining **25%** of the discharge. The **Parman River**, an important tributary, joins the **Phulhar (Jhaua) branch** on its **right bank**, **not** the Barsoi branch.

Q 69. Ans. B

Explanation:

According to the **Bihar Economic Survey 2025–26**, Bihar's economy recorded a **Real GSDP growth rate of 8.6% in 2024–25**, which was **higher than India's estimated Real GDP growth rate of 6.5%** for the same period. Thus, Bihar's economy grew faster than the national average in terms of **real (constant-price) GSDP growth**.

Q 70. Ans. C

Explanation:

- According to the **Bihar Economic Survey 2025–26**, the **Tertiary (Services) Sector** continues to be the **largest contributor to Bihar's Gross State Domestic Product (GSDP)**. It includes activities such as **trade, transport, communication, financial services, real estate, public administration, education, health, and other services**, accounting for **more than half of the State's GSDP**.
- The **Secondary Sector** (manufacturing, construction, electricity, gas, and water supply) has expanded in recent years, particularly due to growth in construction, but **its share is lower than that of the services sector**.
- Mining contributes only a **very small share** to Bihar's economy and is not the dominant sector.

Q 71. Ans. C

Explanation:

- The **Bihar Budget 2026–27** is aligned with the long-term objective of **"Viksit Bihar 2047"**, which complements the national vision of **"Viksit Bharat @2047"**. The budget emphasises sustained economic growth, infrastructure development, quality education, healthcare, employment generation, agricultural modernisation, women's empowerment, and good governance to transform Bihar into a developed state by **2047**.
- The **Bihar Budget 2026–27** highlights that government policies and sectoral allocations are designed to support the vision of **"Viksit Bihar 2047"**, focusing on **inclusive growth, infrastructure, human capital development, industrialisation, and social welfare**.
- While digital governance and e-governance initiatives are part of the state's development strategy, **"Digital Bihar 2035"** is **not** the guiding vision of the **Bihar Budget 2026–27**.

Q 72. Ans. B

Explanation:

- The **Vishwa Shanti Stupa (World Peace Pagoda)** is located on **Ratnagiri Hill** in **Rajgir**, Nalanda district, Bihar. It is one of the most prominent Buddhist pilgrimage sites in India and is a major tourist attraction promoted by the **Department of Tourism, Government of Bihar**.
- The stupa was **inaugurated in 1969** and was built by the **Nipponzan Myohoji**, a Japanese Buddhist order founded by **Nichidatsu Fujii**, with the objective of promoting **world peace and non-violence**. It is one of the several Peace Pagodas constructed by the organisation across different countries.
- **The white marble stupa stands at an elevation of about 400 metres above sea level and can be reached either by a ropeway (one of the oldest aerial ropeways in India) or by a flight of steps. The stupa contains four golden statues of Lord Buddha, depicting the four major events of his life:**
  - ✓ Birth at Lumbini
  - ✓ Enlightenment at Bodh Gaya
  - ✓ First Sermon at Sarnath
  - ✓ Mahaparinirvana at Kushinagar

Q 73. Ans. D

Explanation:

- All three places are important centres of **Jainism** in Bihar and are officially promoted under the **Jain Circuit**
- Pawapuri, in **Nalanda district**, is one of the holiest pilgrimage sites of Jainism. It is the place where **Lord Mahavira**, the **24th Tirthankara**, attained **Mahaparinirvana (Nirvana)** around **527 BCE**. The famous **Jal Mandir**, built in the middle of a lotus-filled pond, marks the cremation site of Lord Mahavira. Every year, thousands of Jain pilgrims visit Pawapuri, especially during **Deepawali**, which commemorates Mahavira's Nirvana.
- Rajgir (ancient **Rajagriha**) is an important pilgrimage centre for both **Jainism and Buddhism**. According to Jain traditions, **Lord Mahavira spent several rainy seasons (Chaturmas)** at Rajgir and delivered many sermons here. The hills around Rajgir contain several **Jain temples** dedicated to different Tirthankaras. Rajgir is also associated with the **20th Tirthankara Munisuvrata**, who is believed to have had connections with the region according to Jain traditions.
- Kundalpur, near Nalanda, is traditionally regarded by the **Digambara Jain tradition** as the **birthplace of Lord Mahavira**. It is one of the most sacred Jain pilgrimage sites in Bihar. The site contains ancient Jain temples and remains an important destination for Jain devotees.

Q 74. Ans. A

Explanation:

The **Tomb of Sher Shah Suri** is situated at **Sasaram** in **Rohtas district, Bihar**. It was built between **1540 and 1545 CE** by **Alawal Khan**, the chief architect of Sher Shah Suri. The mausoleum stands on a **square artificial island** in the middle of a large **man-made water tank (lake)**. A stone causeway connects the tomb to the mainland. This unique architectural setting, combining Afghan and Indo-Islamic styles, makes it one of the finest mausoleums of medieval India. The monument is protected by the **Archaeological Survey of India (ASI)** and is included in the **UNESCO Tentative List**.

Q 75. Ans. A

Explanation:

- **Statement 1 is correct: Calcareous soils (Calcareous Alluvial Soils)** are found mainly in the **north-western part of Bihar**, particularly in districts such as **West Champaran, East Champaran, Gopalganj, Siwan, and parts of Saran**. These soils contain a high proportion of **calcium carbonate (lime)** deposited by Himalayan rivers. They are generally fertile and suitable for crops like **sugarcane, maize, wheat, tobacco, and pulses**, provided proper nutrient management is followed.
- **Statement 2 is incorrect:** Calcareous soils are **not acidic**. Due to their **high calcium carbonate (CaCOβ) content**, they are generally **neutral to moderately alkaline (high pH)**. In Bihar, these soils often exhibit **alkaline characteristics**, which can reduce the availability of micronutrients such as **iron and zinc**. Therefore, describing them as **acidic** is incorrect.

Q 76. Ans. B

Explanation:

During the **medieval period**, **Bihar Sharif** emerged as an important centre for **paper manufacturing**. The town was renowned for producing high-quality handmade paper, which was widely used for **official records, manuscripts, and literary works**. The flourishing paper industry was supported by Bihar Sharif's status as a centre of Islamic learning and administration under the Delhi Sultanate and later the Mughal Empire.

Q 77. Ans. C

Explanation:

- During the **Mughal period**, the **Pargana** was the **principal unit for land revenue administration and collection**. Each pargana comprised several villages and had officials such as the **Shiqdar** (law and order), **Amil/Amin** (revenue collection), **Qanungo** (maintenance of land records), and **Patwari** (village records). Revenue assessment and collection from cultivators were organised at the **pargana level**, making it the key fiscal unit in medieval Bihar.
- A **Suba** was the **largest provincial administrative unit** of the Mughal Empire. Bihar became the **Suba of Bihar** under Emperor Akbar. Although the Suba supervised provincial administration and finance, the actual collection of land revenue was carried out through **parganas**.
  - ✓ Administrative Hierarchy (Mughal Period):
    - Suba → Sarkar → Pargana → Village

Q 78. Ans. B

Explanation:

During the medieval period, the **expansion of agriculture in Bihar was primarily achieved by**

clearing forests and reclaiming wastelands. As the population increased and demand for agricultural produce grew, new lands were brought under cultivation with the encouragement of local rulers, zamindars, and the Mughal administration. This process significantly increased the cultivated area and agricultural output.

Q 79. Ans. B

Explanation:

- **Statement 1 is correct: Tal soils** occur in **low-lying depressions** where rainwater and floodwater accumulate during the southwest monsoon. Owing to poor natural drainage, these soils **remain waterlogged for several weeks or even months**, making kharif cultivation difficult in many areas. After the water recedes, the soils retain sufficient moisture, making them highly suitable for **rabi crops**, particularly **wheat, pulses, gram, lentil, and oilseeds**.
- **Statement 2 is incorrect:** Tal soils are not found along the Indo-Nepal border (that region is characterised by Terai or Piedmont Swamp soils). Instead, Tal soils are found in **southern Bihar**. They are confined to a narrow, 8 to 10-kilometre-wide continuous belt directly along the **southern bank of the river Ganga**, stretching horizontally from Buxar in the west through Patna and Mokama, down to Bhagalpur in the east.

Q 80. Ans. D

Explanation:

In **1906**, **Dr. Rajendra Prasad** served as the **Secretary of the Bihari Club** in **Calcutta**, a debating society established for students from Bihar pursuing their education there. During a meeting of the club on **29 July 1906**, he proposed convening a conference of Bihari students during the forthcoming Puja holidays. Acting on this resolution, he organised the **first session of the Bihari Students' Conference** at **Patna College Hall** in **1906**. The conference emerged as the **first student organisation of its kind in India** and played a significant role in generating public opinion for the **separation of Bihar from the Bengal Presidency**, making it an important milestone in Bihar's political and educational history.

Q 81. Ans. C

Explanation:

- **Assertion (A) is true:** The **Fourth Agriculture Road Map (2023–2028)** of the **Government of Bihar** places significant emphasis on **climate-resilient agriculture**. It promotes measures such as **climate-resilient crop varieties, water-use efficiency, micro-irrigation, crop diversification, soil health management, conservation agriculture, integrated farming systems, and climate-smart agricultural practices** to enhance farmers' resilience against floods, droughts, heat stress, and other climatic risks.
- **Reason (R) is False:** The statement is incorrect because **increasing climate variability has not improved agricultural conditions**. Instead, Bihar faces **greater uncertainty in rainfall, frequent floods in north Bihar, droughts in south Bihar, heat waves, cold waves, and erratic monsoon behaviour**, all of which adversely affect crop productivity and water availability. The emphasis on climate-resilient agriculture is a response to these challenges, not because climate variability has improved agricultural conditions.

Q 82. Ans. C

Explanation:

The **VB-G RAM G (Viksit Bharat – Guarantee for Rozgar and Ajeevika Mission) Act** came into effect across **rural India on 1 July 2026**, replacing the **Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA)**. The new legislation enhances rural livelihood support by increasing the guaranteed employment entitlement from **100 days to 125 days per rural household annually**. It also raises the **minimum daily wage to ₹300**, with the average wage across states estimated at **around ₹327 per day**, compared to the previous wage floor of **₹241**. The Act aims to strengthen employment opportunities and improve income security for rural households.

Q 83. Ans. A

Explanation:

Kapil Dev Prasad was a distinguished master weaver from **Bihar** who played a pivotal role in preserving and promoting the ancient **Bawan Buti** weaving tradition, a textile art closely associated with Bihar's Buddhist heritage. In recognition of his outstanding contribution to India's handloom sector, he was awarded the **Padma Shri**, the country's **fourth-highest civilian honour**, in **2023**. He passed away in **March 2024** at the age of **71** due to heart-related ailments. **Bawan Buti**, which literally means **"52 motifs,"** is a traditional hand-weaving technique characterised by the repeated weaving of **52 identical intricate motifs** on fabric. The designs prominently feature **Buddhist and cultural symbols** such as the **Bodhi**

tree, Dharmachakra (Wheel of Dharma), lotus, bull, fish, conch shell, and trishul. The craft is primarily practised on **cotton and tussar silk**, producing elegant **sarees, bedsheets, curtains, drapes, and other handloom textiles**.

Q 84. Ans. B

Explanation:

Scientists have confirmed that a **banyan tree in Munger, Bihar**, is approximately **700 years old**, making it the **world's oldest accurately dated banyan tree**. Dating back to the **14th century (1300s)**, the tree predates nearby colonial-era structures and has endured through the **Mughal period, British rule, and independent India**. Since tropical trees such as banyans do not develop distinct annual growth rings, researchers from the **Birbal Sahni Institute of Palaeosciences** determined its age using **radiocarbon dating**. The study involved extracting **alpha-cellulose**, a stable component of the plant cell wall, from wood samples collected near the tree's centre. The samples were then analysed using **mass spectrometry** to measure the remaining radioactive carbon, enabling scientists to accurately estimate the tree's age based on the natural decay of carbon over time.

Q 85. Ans. A

Explanation:

- **Assertion (A) is true: Agro-Climatic Zone II (North-East Alluvial Plain)** of Bihar is the principal **jute-growing region** of the state. Districts such as **Katihar, Purnea, Kishanganj, Araria, and Supaul** receive abundant monsoon rainfall and possess fertile alluvial soils, making them highly suitable for jute cultivation. In contrast, **Zone IIIA (South-East Alluvial Plain)** receives comparatively lower rainfall and is better suited for crops such as **paddy, wheat, maize, pulses, and oilseeds** rather than jute.
- **Reason (R) is true:** Zone II receives the **highest annual rainfall in Bihar (generally 1,200–1,700 mm)** due to its proximity to the Himalayan foothills. It is also characterised by **new alluvial soils (Khadar)** deposited by rivers such as the **Kosi and Mahananda.** Jute requires **high rainfall, warm and humid conditions, and fertile alluvial soils**, making Zone II naturally more suitable than Zone IIIA.
- The **higher rainfall and fertile newer alluvial soils** are precisely the reasons why **Zone II has greater potential for jute cultivation**.

Q 86. Ans. A

Explanation:

- **Statement 1 is correct:** The Market Stabilisation Scheme (MSS) was introduced in April 2004 after a Memorandum of Understanding (MoU) was signed between the Reserve Bank of India (RBI) and the Government of India.
- **Statement 2 is correct:** It is a monetary policy intervention tool primarily designed to withdraw (sterilise) excess liquidity from the system. In the early 2000s, massive foreign portfolio investments (capital inflows) forced the RBI to buy dollars and pump rupees into the market. MSS was created to absorb this surplus rupee liquidity and prevent inflation.
- **Statement 3 is incorrect:** This is the most crucial distinction between regular government borrowing and the MSS. The money raised through the auction of MSS bonds or treasury bills is **not** credited to the Consolidated Fund of India and cannot be used by the government to fund its daily expenditures or fiscal deficit. Instead, the proceeds are kept in a separate, identifiable cash account maintained by the RBI (known as the MSS Account) strictly for sterilisation purposes.
- **Additional Facts:**
  - ✓ MSS securities are issued by the RBI on behalf of the Government of India. They are indistinguishable from regular government securities (G-Secs) or Treasury Bills in the secondary market.
  - ✓ Since the government issues these bonds but cannot use the funds, it still has to pay interest on them. The cost of paying this interest (the discount or coupon rate) is borne by the Government of India and is shown as a separate item in the Union Budget.
  - ✓ While initially created for foreign exchange inflows, MSS was heavily utilised during the 2016 demonetisation. When banks were suddenly flooded with massive public deposits of banned ₹500 and ₹1,000 notes, the RBI aggressively issued MSS bonds to mop up this unprecedented surplus liquidity from the banking system.

Q 87. Ans. A

Explanation:

- M3 (Broad Money) is the broadest commonly used measure of money supply in India. It includes all components of M1 (Narrow Money) along with the Net Time Deposits of Commercial Banks.
- M3 = M1 + Net Time Deposits of Commercial Banks

- **Additional Facts:**
  - ✓ M0 (Reserve Money) = Currency in circulation + Bankers' deposits with RBI + Other deposits with RBI
  - ✓ M1 (Narrow Money) = Currency with the public + Demand deposits with banks + Other deposits with RBI
  - ✓ M2 = M1 + Savings deposits with the Post Office Savings System
  - ✓ M3 (Broad Money) = M1 + Net Time Deposits of Commercial Banks
  - ✓ M4 = M3 + Total deposits with the Post Office Savings System (excluding National Savings Certificates)

Q 88. Ans. D

Explanation:

- Participatory Notes (P-Notes) are officially classified by SEBI as Offshore Derivative Instruments (ODIs). They are issued by registered Foreign Portfolio Investors (FPIs) to overseas investors who wish to invest in the Indian financial markets without registering directly with SEBI. The ODI derives its value from an underlying Indian security.
- **According to SEBI regulations governing FPIs and the issuance of ODIs, these instruments can be issued against a wide range of underlying Indian securities:**
  - ✓ **Equity (1):** This is the most common underlying asset. Overseas investors use P-Notes to take positions in Indian listed company shares.
  - ✓ **Debt Securities (2):** FPIs are permitted to invest in Indian government securities (G-Secs), Treasury Bills, and corporate bonds. Consequently, they can issue ODIs with these debt instruments as the underlying asset.
  - ✓ **Hybrid Securities (3):** FPIs are allowed to invest in hybrid instruments such as Real Estate Investment Trusts (REITs) and Infrastructure Investment Trusts (InvITs). P-Notes can therefore be structured with these hybrid securities as the underlying asset.
  - ✓ **Derivatives (4):** P-Notes can have derivatives as the underlying asset, though SEBI strictly regulates this. Following a 2017 SEBI circular, FPIs are restricted from issuing ODIs with derivatives as the underlying unless the derivative positions are taken specifically for hedging the equity shares held by the FPI. Despite this conditional restriction, derivatives remain a legally permissible underlying instrument for P-Notes under specific hedging compliance.

Q 89. Ans. B

Explanation:

- The Debtor Reporting System (DRS) was established by the World Bank in 1951. It is the principal means by which the World Bank captures detailed, loan-by-loan information on the external borrowing of its borrowing member countries.
- **Function and Scope:** Under the DRS, member countries are required to report data on their external public and publicly guaranteed debt, as well as private non-guaranteed debt.
- **Official Output:** The data collected through the DRS is the foundation for the World Bank's annual International Debt Report (formerly known as International Debt Statistics or Global Development Finance). This report provides the most comprehensive and globally consistent dataset on the external debt of low- and middle-income countries.
- **Mandate:** Reporting to the DRS is a condition of World Bank membership and is mandatory for all countries actively borrowing from the International Bank for Reconstruction and Development (IBRD) or the International Development Association (IDA).

Q 90. Ans. B

Explanation:

- **The IMF's Special Drawing Right (SDR) valuation basket currently consists of five currencies:** the U.S. Dollar, Euro, Chinese Renminbi (Yuan), Japanese Yen, and British Pound Sterling. The Chinese Renminbi was added to the basket in 2016, becoming the first currency of an emerging market economy to be included, reflecting China's growing role in global trade and finance.
- The basket composition and currency weights are reviewed by the IMF's Executive Board every five years (the most recent review took effect in 2022, and the next is due around 2027-28), based on criteria like the currency's role in world trade and its status as a "freely usable" currency widely used for international transactions and held as reserves.

Q 91. Ans. C

Explanation:

- **International Finance Corporation (IFC):** The private sector arm of the World Bank Group, established in 1956. It invests directly in private companies and financial institutions in developing countries — through loans, equity investments, and advisory services — without requiring government guarantees, unlike IBRD/IDA lending to sovereign governments. IFC focuses on fostering private-sector-led economic growth, job creation, and sustainable development.

- Additional Facts:
  - ✓ **International Bank for Reconstruction and Development (IBRD):** Lends to middle-income and creditworthy low-income governments — the original "World Bank" institution, established in 1944.
  - ✓ **International Development Association (IDA):** Provides concessional loans and grants to the world's poorest countries — the World Bank's fund for the poorest.
  - ✓ **Multilateral Investment Guarantee Agency (MIGA):** Provides **political risk insurance and credit enhancement** to investors and lenders, encouraging foreign direct investment into developing countries by protecting against risks like expropriation, currency inconvertibility, and political violence.
  - ✓ **International Centre for Settlement of Investment Disputes (ICSID):** Provides a forum for **arbitration and conciliation of investment disputes** between international investors and states.

Q 92. Ans. B

Explanation:

- **The Economic Survey 2025-26 explicitly attributes the sharp decline in rural inflation during 2025 primarily to the easing (disinflation) of food prices. Key findings from the Survey:**
- Unlike 2023 and 2024, when rural inflation exceeded urban inflation (driven by higher food prices), rural inflation in 2025 declined and fell below urban inflation, directly reflecting the sharp fall in food prices.
- Food inflation entered outright deflationary territory from June 2025, with prices of vegetables (especially onions, tomatoes, potatoes), pulses, and spices falling steeply — aided by higher agricultural production, timely trade interventions (export/import management), and strategic buffer stock releases.
- Since rural households spend a disproportionately larger share of their budget on food compared to urban households, this food-price collapse had an outsized moderating effect specifically on rural inflation, offering direct relief to rural consumers.
- Cereal inflation alone fell from 6.2% (January 2025) to -0.4% (December 2025)

Q 93. Ans. D

Explanation:

- **Statement 1 is correct:** The Economic Survey 2025-26 confirms India has emerged as the world's third-largest domestic aviation market, with airports increasing from 74 (2014) to 164 (2025).
- **Statement 2 is correct:** The Survey states India ranks third globally in overall renewable energy and installed solar capacity, with renewables constituting around 49.83% of total power generation capacity as of November 2025.
- **Statement 3 is correct:** Nine Indian ports (Vishakhapatnam plus Pipavav, Kamarajar, Cochin, Hazira, Krishnapatnam, Mundra, Chennai, and JNPA) featured among the World Bank's top 100 in the Container Port Performance Index (CPPI) 2024, confirmed by both the Economic Survey's infrastructure section and the Ministry of Ports, Shipping & Waterways.
- **Statement 4 is incorrect:** The Economic Survey makes no claim that India ranks third in nuclear power capacity. In reality, India's installed nuclear capacity stands at only 8,780 MW, a fraction of leading nuclear nations like the US (~100 GW), France (~61 GW), China, Russia, and South Korea. The Survey instead sets an ambitious target of reaching 100 GW nuclear capacity by 2047 (via the National Nuclear Energy Mission and the SHANTI Act, 2025) — India is nowhere near 3rd globally in current installed nuclear capacity.

Q 94. Ans. C

Explanation:

- **Statement 1 is correct:** Absolute humidity is the measure of the actual, physical amount (mass) of water vapour present in a given volume of air. It represents the true moisture content of the atmosphere, regardless of the air's temperature.
- **Statement 2 is correct:** Because it measures mass per unit volume, absolute humidity is most commonly expressed in **grams of water vapour per cubic meter of air** (g/m^3).
- Additional Facts:
  - ✓ **Relative Humidity:** Unlike absolute humidity, relative humidity is a ratio (expressed as a percentage). It compares the current amount of water vapour in the air to the maximum amount of water vapour the air could possibly

hold at that specific temperature. If the temperature rises but the actual moisture (absolute humidity) stays the same, the relative humidity drops.

- ✓ **Specific Humidity:** This measures the mass of water vapour per unit mass of the total air parcel (usually expressed in grams per kilogram, or g/kg). Meteorologists often prefer specific humidity because, unlike absolute humidity, its value does not change if an air parcel expands or compresses due to changes in pressure or temperature.
- ✓ **The Temperature Limit:** While absolute humidity doesn't depend on temperature to be calculated, temperature dictates the maximum absolute humidity air can reach (the saturation point). Warm air can physically hold significantly more water vapour than cold air before condensation (rain, dew, or fog) occurs.

Q 95. Ans. B

Explanation:

- **Statement 1 is incorrect:** The Gulf of Mexico is a marginal sea of the **Atlantic Ocean**, not the Pacific Ocean. It connects to the Atlantic through the Straits of Florida and to the Caribbean Sea via the Yucatán Channel.
- **Statement 2 is correct:** It is an almost landlocked basin largely surrounded by the North American continent. It is bounded by the southern coast of the United States to the north and east, and by Mexico to the west and south. Cuba borders its southeastern quadrant.
- **Statement 3 is incorrect:** While the Gulf Stream does originate in the Gulf of Mexico, it is a powerful, **warm** ocean current, not a cold one.
- **Additional Facts:**
  - ✓ Inside the Gulf of Mexico, the warm water from the Caribbean flows in a horseshoe-shaped loop (the Loop Current) before exiting through the Straits of Florida to become the Gulf Stream. This deep pool of warm water provides massive amounts of heat energy that frequently fuels and intensifies hurricanes (like Katrina and Harvey).
  - ✓ As the warm Gulf Stream travels up the eastern coast of the US and across the Atlantic towards Europe, it significantly warms the climate of Western Europe, making countries like the UK and Norway much milder than other places at similar latitudes.
  - ✓ **Dead Zones:** The Gulf of Mexico experiences a massive, recurring "dead zone" (hypoxic zone) every summer. This is caused by nutrient runoff (largely agricultural fertilisers) travelling down the Mississippi River basin and pouring into the Gulf, which triggers massive algal blooms that deplete the water of oxygen, killing marine life.

Q 96. Ans. A

Explanation:

- **Statement A is correct:** The defining characteristic of a hot spring (as opposed to a geyser) is that the heated groundwater rises to the surface and discharges quietly and continuously. Because the underground plumbing of a hot spring does not have the narrow constrictions that trap steam and build pressure, the water flows out without any violent eruption.
- **Statement B and C are incorrect:** These statements describe geysers. A geyser has a complex underground structure with narrow bottlenecks. Water deep down gets superheated and turns into steam, creating immense pressure that eventually forces the water above it to erupt violently as a fountain of boiling water and steam.
- **Statement D is incorrect:** While hot springs are incredibly common in active thermal or volcanic regions (like Yellowstone or Iceland), they are not strictly restricted to them. They can also occur in non-volcanic areas where water percolates deep into the Earth's crust along faults, gets heated by the natural geothermal gradient, and is forced back up to the surface.
- **Additional Facts:**
  - ✓ **Fumarole:** A vent in the Earth's surface that emits only steam and volcanic gases (like sulfur dioxide or hydrogen sulfide), but no liquid water. If a hot spring boils away completely, it can become a fumarole.
  - ✓ **Mudpot:** A type of hot spring that is short of water. Acidic gases break down surrounding rock into clay, which mixes with the limited water to form a bubbling, boiling puddle of mud.
  - ✓ **Geysers are Rare:** There are only about 1,000 active geysers worldwide, and more than half of them are located in Yellowstone National Park in the USA. Hot springs, on the other hand, are found all over the globe in almost every country.
  - ✓ India has several famous hot springs, such as Manikaran in Himachal Pradesh, Panamik in

Ladakh, and Bakreshwar in West Bengal, which are often valued for their mineral content and perceived therapeutic properties.

Q 97. Ans. B

Explanation:

- Geysers are powered by Earth's internal (geothermal) heat. In regions of active or recent volcanism, groundwater that seeps down through cracks and fissures comes into contact with hot rocks or magma chambers close to the surface. This heat source superheats the trapped water well above its normal boiling point (due to the immense pressure exerted by the overlying column of water and rock), eventually causing it to flash into steam and erupt violently through a narrow vent as a geyser.
- This is the same source of energy — heat generated by radioactive decay in the Earth's mantle and core, and residual heat from planetary formation — that drives other geothermal phenomena like hot springs, fumaroles, and volcanic activity itself.

Q 98. Ans. B

Explanation:

- **Statement 1 is correct:** The vertical pressure gradient force is indeed much larger than the horizontal pressure gradient force. However, despite its greater magnitude, it does not produce large-scale vertical winds because it is almost exactly counterbalanced.
- **Statement 2 is incorrect:** The vertical pressure gradient force is generally balanced by gravity (the gravitational force), not the Coriolis force. This near-perfect balance between the upward-acting vertical pressure gradient force and the downward-acting force of gravity is what keeps the atmosphere in hydrostatic equilibrium, preventing large net vertical air movement. The Coriolis force, by contrast, acts specifically on horizontal wind motion (deflecting winds to the right in the Northern Hemisphere and left in the Southern Hemisphere), interacting with the horizontal pressure gradient force to produce geostrophic winds, it plays no role in balancing the vertical component.
- **Statement 3 is incorrect:** Precisely because the vertical pressure gradient force is balanced by gravity, it does not cause strong upward winds over most parts of the Earth. Vertical air movement is generally weak and localised (occurring mainly in specific settings like convective thunderstorm cells, monsoon depressions, or frontal systems), not a widespread global phenomenon.

Q 99. Ans. C

Explanation:

- A tsunami is a series of giant ocean waves caused by the sudden displacement of a massive volume of water. The most common trigger is an undersea **earthquake** (usually at tectonic subduction zones) that causes the seafloor to violently thrust upward or drop downward. **Powerful volcanic eruptions** (like the 2022 Hunga Tonga eruption or Krakatoa in 1883) and massive **underwater landslides** (or landslides that fall into the ocean) also displace enough water to generate tsunamis.
- **Tropical cyclones** (hurricanes or typhoons) do not cause tsunamis; they cause **storm surges**. While both result in devastating coastal flooding, their origins are entirely different. A storm surge is a meteorological phenomenon driven by high winds pushing water onto the shore and low atmospheric pressure. A tsunami is a geological phenomenon.
- **Additional Facts:**
  - ✓ **Meteorite Impacts:** While exceedingly rare, the impact of a large meteorite or asteroid into an ocean is another scientifically recognised trigger for a megatsunami.
  - ✓ **Deep vs. Shallow Water (Shoaling):** In the deep ocean, tsunami waves travel at speeds matching commercial jetliners (over 800 km/h) but may only be a few tens of centimetres high, making them practically unnoticeable to ships. As the wave approaches the shallow coastal waters, friction with the seabed slows it down drastically. This compressed energy forces the water column upward, creating the massive, destructive wall of water known as shoaling.
  - ✓ **The Ring of Fire:** About 80% of all tsunamis occur in the Pacific Ocean's "Ring of Fire," a seismically active horseshoe-shaped zone defined by frequent earthquakes and volcanic activity.

Q 100. Ans. C

Explanation:

- **Pair 1 is correctly matched:** The North Atlantic Drift is the eastward continuation of the Gulf Stream, splitting off near 40°N, 30°W and travelling toward Northwest Europe, giving countries like Norway and the British Isles much milder winters than their latitude would otherwise suggest.
- **Pair 2 is correctly matched:** The Brazil Current is a warm current flowing southward along the eastern

(Atlantic) coast of South America, formed as the southern branch of the South Equatorial Current splits near Cape de São Roque, Brazil.

- **Pair 3 is correctly matched:** The Florida Current flows through the Straits of Florida, rounding the Florida Peninsula, where it merges with the Antilles Current to form the initial portion of the Gulf Stream system.
- **Pair 4 is incorrectly matched:** The Antilles Current flows northwestward along the north side of the Greater Antilles / West Indies, in the North Atlantic Ocean — nowhere near the western (Pacific) coast of South America. It is a branch of the North Equatorial Current that merges with the Florida Current to help form the Gulf Stream. The current that actually flows along the western coast of South America is the cold Peru (Humboldt) Current, an entirely different current.

Q 101. Ans. A

Explanation:

- **The Red Sea is a narrow sea located between northeastern Africa and the Arabian Peninsula. It is bordered by six countries:**
  - ✓ Saudi Arabia
  - ✓ Yemen
  - ✓ Egypt
  - ✓ Sudan
  - ✓ Eritrea
  - ✓ Djibouti
- **The Red Sea is connected:**
  - ✓ North → Mediterranean Sea through the Suez Canal.
  - ✓ South → Gulf of Aden and Arabian Sea through the Bab el-Mandeb Strait.

Q 102. Ans. A

Explanation:

- **The "Indian Plate" refers to the major tectonic plate (often grouped as the broader Indo-Australian Plate) that includes the following landmasses:**
  - ✓ **Peninsular India (1):** This is the core continental crust of the Indian Plate. Millions of years ago, it broke off from the ancient supercontinent Gondwana and began a long northward journey, eventually colliding with the Eurasian Plate to form the Himalayas.
  - ✓ **Australian continental portion (2):** The boundaries of the Indian Plate include both Peninsular India and the Australian continental mass, which are bound together dynamically.
- **Additional Facts:**
  - ✓ **Arabian Peninsula (3):** This region rests on an entirely different tectonic plate called the Arabian Plate. It is separated from the Indian Plate by a transform boundary known as the Owen Fracture Zone in the Arabian Sea.
  - ✓ **Madagascar (4):** While India and Madagascar were once fused together in Gondwana, they broke apart roughly 88–90 million years ago. Following this split, the Indian Plate moved rapidly northwards, leaving Madagascar behind. Today, Madagascar is part of the African Plate (specifically on the Somali subplate).

Q 103. Ans. C

Explanation:

- **Pair 1 is correct:** Tamil Nadu — Northeast Monsoon
  - ✓ During the winter months (October to December), the Southwest monsoon retreats and the wind pattern reverses to form the Northeast monsoon (also known as the retreating monsoon). As these northeast trade winds blow from land to sea, they are generally dry. However, they pick up significant moisture as they cross the Bay of Bengal, delivering the bulk of the annual rainfall to the Coromandel Coast, particularly Tamil Nadu and parts of coastal Andhra Pradesh.
- **Pair 2 is correct:** Punjab — Western Disturbances
  - ✓ Winter rainfall in Northwest India is primarily caused by Western Disturbances. These are shallow cyclonic depressions (extratropical storms) that originate over the Mediterranean Sea. They are steered into the Indian subcontinent by the westerly jet streams. Upon reaching India, they cause widespread precipitation in the northern states, including Punjab and Haryana, which is crucial for the winter (Rabi) crops like wheat.
- **Pair 3 is correct:** Bihar — Eastward-moving Western Disturbances
  - ✓ As the Western Disturbances travel eastwards across the Indo-Gangetic plain, they progressively lose moisture. However, they remain active enough to cause light, scattered winter rainfall as far east as Bihar and sometimes even West Bengal. While the total volume of rain is much lower than in Punjab, the mechanism driving this winter precipitation in Bihar is exactly the same eastward-moving system.

Q 104. Ans. A

Explanation:

- **Election of Vice-President (a - 4):** Article 66 outlines that the Vice-President is elected indirectly by an electoral college consisting of members (both elected and nominated) of both Houses of Parliament, using the system of proportional representation by means of the single transferable vote.
- **Oath of office of Vice-President (b - 3):** Article 69 prescribes the oath or affirmation that the Vice-President must make before entering office. It is administered by the President of India or some person appointed in that behalf.
- **Vice-President to act as President (c - 1):** Article 65 states that the Vice-President shall act as President in the event of the occurrence of any vacancy in the office of the President by reason of their death, resignation, removal, or otherwise.
- **Ex officio Chairman of the Rajya Sabha (d - 2):** Article 64 declares that the Vice-President shall be the ex officio Chairman of the Council of States (Rajya Sabha) and shall not hold any other office of profit.
- **Additional Facts:**
  - ✓ **Salary:** The Constitution has not fixed any salary for the Vice-President in that specific capacity. They draw their regular salary in their role as the ex officio Chairman of the Rajya Sabha. If they step in to act as President under Article 65, they draw the salary and allowances of the President instead.
  - ✓ **Removal:** The Vice-President can be removed from office by a resolution of the Rajya Sabha passed by an effective majority (a majority of all the then members) and agreed to by the Lok Sabha with a simple majority. No formal impeachment process is required, like it is for the President.
  - ✓ **Disputes:** According to Article 71, all doubts and disputes arising out of or in connection with the election of a Vice-President are inquired into and decided by the **Supreme Court of India**, whose decision is final.

Q 105. Ans. C

Explanation:

- **NHRC (Statutory body) - Correctly matched:** The National Human Rights Commission was established in 1993 under the Protection of Human Rights Act, 1993. Since it was created by an Act of Parliament, it is a statutory body, not a constitutional one.
- **State Human Rights Commission (Statutory body) - Correctly matched:** Like the NHRC, State Human Rights Commissions are also created under the authority of the same parliamentary law—the Protection of Human Rights Act, 1993.
- **National Commission for Women (Constitutional body) - Incorrectly matched:** The NCW is a **statutory body**, not a constitutional body. It was set up in 1992 under the National Commission for Women Act, 1990, to review constitutional and legal safeguards for women. It is not mentioned in the Constitution of India.
- **Central Vigilance Commission (Statutory body since 2003) - Correctly matched:** The CVC was originally set up in 1964 through a simple executive resolution of the Government of India. However, it was later conferred statutory status by the enactment of the Central Vigilance Commission Act, 2003.
- **Additional Facts:**
  - ✓ **Constitutional Bodies:** These derive their powers directly from the Constitution of India. They have dedicated articles. Examples include the Election Commission of India (Article 324), the Finance Commission (Article 280), and the National Commission for SCs (Article 338). Altering their mandate requires a formal constitutional amendment.
  - ✓ **Statutory Bodies:** These are created by an Act of Parliament or a State Legislature. They derive their powers from the specific statute (law) passed to create them. Parliament can alter their powers or dissolve them simply by amending or repealing the act.
  - ✓ **The Santhanam Committee:** The Central Vigilance Commission (CVC) was originally established based on the recommendations of the Committee on Prevention of Corruption, headed by K. Santhanam. The CVC is conceived to be the apex vigilance institution, free of control from any executive authority.

Q 106. Ans. A

Explanation:

- **Pair 1 is correctly matched:** The Ninth Schedule contains a list of central and state laws (mostly regarding land reforms and abolition of the zamindari system) which cannot be challenged in courts on the ground of violation of fundamental rights.
- **Pair 2 is correctly matched:** The Tenth Schedule contains provisions regarding the disqualification

of members of Parliament and State Legislatures on the ground of defection (commonly known as the Anti-Defection Law).

- **Pair 3 is incorrectly matched:** The Eleventh Schedule specifies the powers, authority, and responsibilities of **Panchayats** (rural local government), not Municipalities. It contains 29 functional items.
- **Pair 4 is incorrectly matched:** The Twelfth Schedule specifies the powers, authority, and responsibilities of **Municipalities** (urban local government), not Panchayats. It contains 18 functional items.
- **Additional Facts:**
  - ✓ **Original Constitution:** When the Constitution of India was adopted in 1949, it originally contained only **eight** schedules. Schedules 9 through 12 were added later through various constitutional amendments.
  - ✓ **First Amendment (1951):** The Ninth Schedule was added by the very first amendment to protect agricultural land reforms from intense judicial scrutiny. (Note: In 2007, the Supreme Court ruled that laws placed in the 9th Schedule after April 24, 1973, are still open to judicial review if they violate the "basic structure" of the Constitution.
  - ✓ **Fifty-Second Amendment (1985):** The Tenth Schedule was added during Rajiv Gandhi's tenure to curb the political menace of "horse-trading" and frequent party-hopping by elected representatives.
  - ✓ **The 1992 Amendments:** The 73rd and 74th Constitutional Amendments fundamentally changed democratic decentralisation in India by formally recognising local self-governments and adding the 11th and 12th Schedules, respectively.

Q 107. Ans. A

Explanation:

- **Saving of laws providing for the acquisition of estates (a - 2):** Article 31A protects laws relating to agricultural land reforms, the acquisition of estates, and the taking over of management of properties from being challenged on the grounds that they violate the Fundamental Rights guaranteed by Article 14 (equality) and Article 19 (freedoms).
- **Validation of Acts placed in the Ninth Schedule (b - 3):** Article 31B acts as a constitutional shield. It explicitly states that none of the acts and regulations placed in the Ninth Schedule shall be deemed void on the ground that they are inconsistent with or take away any of the Fundamental Rights.
- **Saving of laws giving effect to certain Directive Principles (c - 1):** Article 31C protects any law enacted to implement the socialist Directive Principles specified in Article 39(b) and (c) (which deal with equitable distribution of material resources and preventing the concentration of wealth) from being challenged on the grounds of violating Articles 14 and 19.
- **Bar to interference by courts in electoral matters (d - 4):** Article 329 restricts the judiciary from interfering in matters relating to the delimitation of constituencies or the allotment of seats. It also dictates that no election to Parliament or a State Legislature can be called into question except by an election petition presented to the appropriate authority.
- **Additional Facts:**
  - ✓ **The First Amendment (1951):** Articles 31A and 31B, along with the Ninth Schedule, were introduced by the First Constitutional Amendment. The primary goal was to overcome judicial hurdles that were preventing the newly formed government from abolishing the Zamindari system and implementing sweeping land reforms.
  - ✓ **The Twenty-Fifth Amendment (1971):** Article 31C was inserted later by the 25th Amendment during Indira Gandhi's tenure. It marked a significant shift in constitutional interpretation because it was the first time that specific Directive Principles (which are generally non-justiciable) were given explicit primacy over Fundamental Rights.
  - ✓ **The Basic Structure Doctrine:** While Article 31B was designed as a blanket shield, the Supreme Court ruled in the landmark I.R. Coelho case (2007) that any laws placed in the Ninth Schedule after April 24, 1973 (the date of the Kesavananda Bharati judgment), are still open to judicial review if they violate the "basic structure" of the Constitution.

Q 108. Ans. C

Explanation:

- **Mandamus (Incorrectly matched):** The writ of Mandamus (meaning "We command") is issued by a court to a public official, government, or lower court, asking them to perform the official public duties that they have failed or refused to perform. It **cannot** be issued against a private individual or a private body (unless they are entrusted with a public duty).

- **Habeas corpus (Correctly matched):** The writ of Habeas corpus (meaning "to have the body of") is an order to produce a detained person before the court. It is unique because it can be issued against **both** public authorities and private individuals who have illegally detained someone.
- **Quo warranto (Correctly matched):** The writ of Quo warranto (meaning "by what authority") is issued to inquire into the legality of a person's claim to a public office. Unlike other writs where the strict rule of locus standi applies (meaning only the aggrieved person can approach the court), Quo warranto can be sought by **any interested or public-minded person**, not necessarily the aggrieved party.
- **Certiorari (Correctly matched):** The writ of Certiorari (meaning "to be certified") is issued by a higher court to a lower court or tribunal either to transfer a case or to **quash their order** in a case, usually on the grounds of excess of jurisdiction or error of law.
- Therefore, three out of the four pairs are correctly matched.
- **Additional Facts:**
  - ✓ **Constitutional Basis:** Writs are issued by the Supreme Court under Article 32 (for the enforcement of Fundamental Rights only) and by the High Courts under Article 226 (for Fundamental Rights as well as for any other ordinary legal right).
  - ✓ **Prohibition vs. Certiorari:** Both writs are issued against lower courts or tribunals, but they differ in timing. Prohibition is issued to **prevent** a lower court from exceeding its jurisdiction (issued during the proceedings), whereas Certiorari is issued to **quash** an order that has already been passed by the lower court. "Prevention is better than cure"—Prohibition is prevention; Certiorari is the cure.
  - ✓ **Borrowing:** The concept of prerogative writs in the Indian Constitution is borrowed from English common law, where they were known as the "fountain of justice."

Q 109. Ans. A

Explanation:

- According to **Section 151A** of the Representation of the People Act, 1951, a by-election to fill a casual vacancy in either the House of Parliament or a State Legislature must be held within **six months** from the date of the occurrence of the vacancy.
- **However, the Act specifies two clear exceptions where this requirement does not apply:**
  - ✓ **The One-Year Rule:** If the remainder of the term of the member in relation to the vacancy is **less than one year**, the by-election does not need to be held. The rationale is that the administrative cost and effort of conducting an election are disproportionate to the very short period of service remaining for the new member.
  - ✓ **Impracticality/Difficulty:** If the Election Commission, in consultation with the Central Government, certifies that it is difficult to hold the by-election within the specified six-month period due to circumstances that make conducting the poll impractical.
- **Additional Facts:**
  - ✓ **Casual Vacancy Definition:** A casual vacancy arises due to the death, resignation, or disqualification of a sitting member of the House.
  - ✓ **Discretion:** While the Election Commission is the constitutional body responsible for conducting these elections, it is bound by these statutory provisions and cannot arbitrarily choose to skip a by-election unless one of the conditions above is met.

Q 110. Ans. B

Explanation:

- **Appointment of Governor (Article 155):** Correct. The Governor of a state is appointed by the President by warrant under his hand and seal.
- **Qualifications for appointment as Governor (Article 157):** Correct. The two essential qualifications are that the person must be a citizen of India and must have completed the age of 35 years.
- **Conditions of the Governor's office (Article 158):** Correct. This article lays down that the Governor shall not be a member of either House of Parliament or a House of the Legislature of any State, and shall not hold any other office of profit.
- **Oath or affirmation by the Governor (Article 160): Incorrect.** Article 160 deals with the "Discharge of the functions of the Governor in certain contingencies" (e.g., when the Governor dies or is unable to perform duties). The **Oath or affirmation by the Governor** is actually prescribed under **Article 159**, where the Governor swears to preserve, protect, and defend the Constitution and the law.
- **Additional Facts:**
  - ✓ **Article 153:** States that there shall be a Governor

for each state. A constitutional amendment passed in 1956 facilitates the appointment of the same person as Governor for two or more states.
  - ✓ **Article 154:** Executive power of the State is vested in the Governor and is exercised by him either directly or through officers subordinate to him in accordance with the Constitution.
  - ✓ **Article 161:** Grants the Governor the power to grant pardons, reprieves, respites, or remissions of punishment or to suspend, remit, or commute the sentence of any person convicted of any offence against any law relating to a matter to which the executive power of the state extends.
  - ✓ **Article 163:** Mentions that there shall be a Council of Ministers with the Chief Minister at the head to aid and advise the Governor, except in cases where the Governor is required to act in his discretion.

Q 111. Ans. C

Explanation:

- **Whitley Award 2026 (a – 2):** Parveen Shaikh was awarded the 2026 Whitley Award for her exceptional community-based conservation work to protect the endangered Indian skimmer bird.
- **Business Reformer of the Year 2025 (b – 1):** N. Chandrababu Naidu is widely recognised for his extensive initiatives and reforms in the business and infrastructure sectors, particularly in the context of the "Ease of Doing Business" initiatives often cited in recent governance honours.
- **Metro Woman of the Year (c – 4):** This title is awarded to women who have made significant contributions to the development or operation of metro infrastructure, with Sushma Yadav being the recipient in the context of this specific pairing.
- **Global Green Icon (d – 3):** Former Rajya Sabha MP Joginipally Santosh Kumar was honoured with the "Global Green Icon" award at the World Climate Leaders' Summit held in April 2026 for his environmental leadership.

Q 112. Ans. C

Explanation:

- During the operational fiscal year 2025–26, India's Major Ports (under the Ministry of Ports, Shipping and Waterways) collectively handled **915.17 million tonnes (MT)** of cargo. This performance comfortably surpassed the annual target of 904 MT and represented a strong year-on-year growth rate of **7.06 percent**.
- **Additional Facts:**
  - ✓ **Top Performing Ports (2025-26):** The **Deendayal Port Authority** led the country in total cargo handling (160.11 MT), closely followed by the **Paradip Port Authority** (156.45 MT) and the **Jawaharlal Nehru Port Authority (JNPA)** (102.01 MT).
  - ✓ **Highest Growth:** Among the major ports, the **Mormugao Port Authority** recorded the highest individual year-on-year growth rate at 15.91%, followed by the Kolkata Dock System (14.28%).
  - ✓ **Strategic Drivers:** The record throughput was largely attributed to capacity augmentation, infrastructure modernization under the Maritime Amrit Kaal Vision 2047, improved multimodal connectivity, and the adoption of digital and smart port initiatives.

Q 113. Ans. B

Explanation:

- The term **"Sovereign AI"** refers to a nation's capability and strategy to develop, deploy, and govern Artificial Intelligence using its own domestic infrastructure, data, and computational resources.
- As AI becomes a critical general-purpose technology, countries are increasingly viewing it as a matter of national security, economic independence, and cultural preservation. The goal of Sovereign AI is to prevent reliance on foreign tech monopolies for critical technological infrastructure. By building state-backed compute capacity and utilising domestic datasets, a country ensures that its AI models accurately reflect its own laws, culture, languages, and values, while keeping sensitive data within its borders.
- **Additional Facts:**
  - ✓ **The Three Pillars:** True Sovereign AI generally rests on three main pillars: indigenous compute capacity (domestic data centres and GPUs), sovereign data (controlling, securing, and utilising citizen data locally), and a skilled domestic workforce/startup ecosystem.
  - ✓ **India's Approach:** The Government of India has heavily prioritised Sovereign AI through the **IndiaAI Mission** (approved in early 2024 with an outlay of ₹10,371 crore). A major component of this mission involves establishing a robust domestic AI compute infrastructure, comprising over 10,000 GPUs, to offer democratised, subsidised access to Indian startups, academia, and researchers.

- ✓ **Linguistic Sovereignty:** Initiatives like **Bhashini** (India's AI-led language translation platform) are a core part of the Sovereign AI push. They ensure that next-generation Large Language Models (LLMs) understand and serve India's diverse regional languages, rather than being fundamentally biased toward English and Western socio-cultural contexts.

Q 114. Ans. A

Explanation:

- Mission MITRA (Mapping of Interoperable Traits and Response Assessment) is a first-of-its-kind high-altitude analog mission launched by the Indian Space Research Organisation (ISRO) in **Leh**, Ladakh.
- **Objective:** The mission utilises the region's freezing temperatures, low oxygen (hypoxia), and extreme isolation to act as a natural, terrestrial analog for spaceflight operations.
- **Significance:** It is designed to test the physiological dynamics, psychological resilience, and team interoperability of the designated Gaganyaan astronauts (Gaganyatris) and ground control teams before undertaking actual human spaceflight.

Q 115. Ans. B

Explanation:

- **Eri silk processing unit (a - 3):** In May 2026, the Chief Minister of **Meghalaya** inaugurated an Eri silk (locally known as Ryndia) processing unit in the Ri-Bhoi district to strengthen the state's traditional silk sector and provide better market access for local weavers.
- **Ginger Mission (b - 4):** The government of **Mizoram**, along with the Ministry of Development of North Eastern Region (MDoNER), launched a ₹189.79-crore Ginger Mission in mid-2026. The goal is to transform the state's Geographical Indication (GI)-certified, pharma-grade "Mizo ginger" into a globally recognised premium product.
- **Vibrant Villages Programme-II (c - 1):** The Vibrant Villages Programme (VVP) aims to develop border villages to stem migration and bolster national security. In early 2026, the J&K administration systematically extended VVP-II to include strategic border villages in the Kathua district of **Jammu and Kashmir**.
- **Kalai-II project (d - 2):** The Kalai-II Hydroelectric Project is a massive 1200 MW run-of-the-river power project located on the Lohit River in the Anjaw district of **Arunachal Pradesh**.

Q 116. Ans. D

Explanation:

- **Water Forward initiative for water security (The World Bank) - Correctly matched:** Launched by the World Bank Group in April 2026, the 'Water Forward' initiative is a global multistakeholder platform designed to improve water security for 1 billion people by 2030. It focuses on transforming water systems from a source of risk into a driver of economic growth and job creation.
- **India's first semi-high-speed rail project (Ahmedabad) - Correctly matched:** In May 2026, the Union Cabinet approved India's first semi-high-speed double-line rail project. The 134-km corridor uses indigenous technology and connects Sarkhej (Ahmedabad) with the Dholera Special Investment Region (SIR) in Gujarat.
- **India's first pearl farming cluster (Hazaribagh) - Correctly matched:** Hazaribagh in Jharkhand has been officially developed as India's first and only freshwater pearl farming cluster under the Pradhan Mantri Matsya Sampada Yojana (PMMSY). The project aims to utilise village ponds and reservoirs to boost rural livelihoods and women-led enterprises.
- **Landed the reusable booster of the New Glenn rocket (Blue Origin) - Correctly matched:** Jeff Bezos' space firm Blue Origin successfully achieved the vertical sea-platform landing of its heavy-lift New Glenn rocket's first-stage reusable booster, joining a highly exclusive club of aerospace organisations capable of orbital rocket recovery.

Q 117. Ans. C

Explanation:

- **Statement 1 is correct: Project Seabird** (which houses INS Kadamba at Karwar, Karnataka) is currently being executed as the largest naval infrastructure project in India and is widely slated to become Asia's largest naval base upon completion of its expansion phases. The base will provide the Indian Navy with massive strategic depth and operational flexibility on the western seaboard.
- **Statement 2 is correct:** A defining feature of INS Kadamba is its state-of-the-art **Shiplift and Ship Transfer System** (Syncrolift). This specialised facility (originally with a 10,000-tonne capacity) allows the Navy to lift large frontline warships (capital ships like the Project 15B destroyers and Project 17A stealth frigates) as well as submarines directly out of the water for simultaneous dry-docking, repair, and maintenance.

- **Additional Facts (Project Seabird Phase IIA):**
  - ✓ **Strategic Location:** Located south of Mumbai, Karwar provides a deep-water natural harbour that is less congested than Mumbai. Crucially, it is out of range for most strike aircraft from neighbouring adversarial countries, offering a safer staging ground for the Western Fleet.
  - ✓ **The Covered Dry Berth:** The centrepiece of the Phase IIA expansion is a massive, iconic Covered Dry Berth. Standing at 75 meters tall (taller than the Qutub Minar), it is designed to facilitate the simultaneous docking and comprehensive enclosed maintenance of up to four capital ships.
  - ✓ **Aircraft Carrier Berthing:** While the shiplift system handles destroyers and submarines, Phase II expansions include massively reinforced piers specifically designed to berth India's aircraft carriers (INS Vikramaditya and the indigenously built INS Vikrant).

Q 118. Ans. C

Explanation:

- Akasa Air recently became the **fifth** Indian airline to officially join the International Air Transport Association (IATA), the global trade association representing over 360 carriers worldwide.
- **Before Akasa Air's induction, the four other Indian carriers holding IATA membership were:**
  - ✓ Air India
  - ✓ Air India Express
  - ✓ IndiGo
  - ✓ SpiceJet
- **Additional Facts:**
  - ✓ **The Safety Prerequisite:** Akasa Air gained admission into IATA (in January 2026) after successfully clearing the IATA Operational Safety Audit (IOSA). This is a stringent, internationally recognised safety audit that is a mandatory prerequisite for any airline seeking IATA membership.
  - ✓ **Significance:** IATA members account for more than 80% of global air traffic. Being a member gives Akasa Air a seat at the table to participate in global industry dialogues, collaborate on international aviation standards (like digital transformation and sustainability), and significantly boosts the airline's global credibility as it expands its international flight footprint.
  - ✓ **Rapid Growth:** Founded in 2020 and heavily backed by the late investor Rakesh Jhunjhunwala, Akasa Air is India's youngest airline but it achieved this milestone in just over three years of commencing operations.

Q 119. Ans. C

Explanation:

- Financial Software and Systems (FSS) became the first payments company in India, the Middle East, Asia-Pacific (APAC), and South Africa to receive the prestigious **ISO/IEC 42001 certification** in January 2026.
- **Additional Facts:**
  - ✓ **What is ISO/IEC 42001?** Developed by the International Organisation for Standardisation (ISO), it is the world's first global standard for **Artificial Intelligence Management Systems (AIMS).**
  - ✓ **Significance:** This certification signifies a transition from ad-hoc AI adoption to a structured, regulated, and enterprise-wide AI governance framework. It formally recognises FSS's responsible, ethical, and secure deployment of Artificial Intelligence in its payment processing operations, fraud detection, and transaction monitoring.

Q 120. Ans. B

Explanation:

- **Statement 1 is incorrect:** In May 2026, the **Indian Institute of Technology Madras (IIT Madras)** launched its first international centre in Menlo Park, California, United States. Established by the IITM Global Research Foundation, this $7.5 million deep-tech hub serves as a direct launchpad for Indian deep-tech startups to access Silicon Valley's venture capital, mentorship networks, and global markets.
- **Statement 2 is correct: Karnataka** recently launched India's first state-led Centre of Excellence (CoE) in Space Technology in **Bengaluru.** Formalised through a Memorandum of Agreement (MoA) between the Karnataka Innovation and Technology Society (KITS) and the SatCom Industry Association-India (SIA-India), the CoE aims to incubate space-tech startups, fund prototypes, and drive R&D in satellite components and AI-based analytics.
- **Additional Facts:**
  - ✓ The Menlo Park centre represents a shift in India's academic strategy, moving from just exporting talent to actively taking indigenous

research, startups, and emerging technologies directly to the global stage. IIT Madras has also announced plans for a second centre on the US East Coast.

- ✓ With legacy institutions like ISRO, IISc, and DRDO headquartered in Bengaluru, Karnataka already forms the backbone of India's aerospace sector. Through this new CoE (guided by a 2025–2030 roadmap), the state government officially aims to capture 50% of India's projected ₹1.8 lakh crore space market by 2033.

Q 121. Ans. A

Explanation:

- **Best Director: Aditya Dhar** won the Chetak Screen Award 2026 for Best Director for his espionage thriller Dhurandhar.
- **Best Actor:** The Chetak Screen Award 2026 for Best Actor (Male) was won by **Ranveer Singh** for his performance in Dhurandhar, not "Ranbir Singh" (which is a deliberate name mix-up between Ranbir Kapoor and Ranveer Singh).
- **Best Actress:** Yami Gautam Dhar bagged the Best Actor (Female) award for her acclaimed performance in the film Haq.

Q 122. Ans. C

Explanation:

- The HEALD initiative was launched by the Union Home Minister to tackle **liver** diseases on a nationwide scale.
- Concurrently, to support this mission, India's first Integrated Liver Habilitation Centre was inaugurated at the Institute of Liver and Biliary Sciences (ILBS) in New Delhi.

Q 123. Ans. B

Explanation:

- **Statement 1 is incorrect:** The Subhash Chandra Bose Aapda Prabandhan Puraskar 2026 in the Institutional Category was awarded to the **Sikkim State Disaster Management Authority (SSDMA)**, not the National Institute of Disaster Management (NIDM). SSDMA was recognised for its pioneering work in community-centred disaster resilience and deploying trained Aapda Mitras across the state.
- **Statement 2 is correct:** The award in the Individual Category for 2026 was given to **Lieutenant Colonel Seeta Ashok Shelke**. She gained national acclaim for leading Humanitarian Assistance and Disaster Relief (HADR) operations during the devastating 2024 Wayanad (Kerala) landslides, notably supervising the rapid construction of a 190-foot Bailey Bridge in challenging terrain.
- **Additional Facts:**
  - ✓ **Announcement:** The award is announced annually on **January 23rd**, which is the birth anniversary of Netaji Subhash Chandra Bose (celebrated as Parakram Diwas).
  - ✓ **Objective:** Instituted by the Government of India (under the Ministry of Home Affairs / NDMA), it honours the invaluable contribution and selfless service rendered by individuals and organisations in India in the field of disaster management.
  - ✓ **Prize Money:** The award carries a cash prize of **₹51 lakh and a certificate for an institution** (the money must be utilised for disaster management activities only), and **₹5 lakh and a certificate for an individual**.

Q 124. Ans. D

Explanation:

- The Asola Bhatti Wildlife Sanctuary is located on the **Southern Delhi Ridge** of the **Aravalli hill** range, straddling the border between Delhi and Haryana (specifically Faridabad and Gurugram districts). It has frequently been in the news regarding the demarcation and enforcement of its Eco-Sensitive Zone (ESZ) to protect its fragile ecosystem from rampant urban encroachment and illegal mining activities in the National Capital Region (NCR).
- **Additional Facts:**
  - ✓ **Ecological Significance:** The sanctuary acts as a crucial "green lung" for the heavily polluted Delhi-NCR region and serves as a natural barrier preventing the advancement of desertification from the Thar Desert in Rajasthan.
  - ✓ **Wildlife Corridor:** It is a vital part of the Northern Aravalli leopard wildlife corridor, which extends from Sariska National Park in Rajasthan to the Delhi Ridge. Over recent years, camera traps have confirmed the returning presence of apex predators like leopards in the sanctuary.
  - ✓ **Historical Context:** The "Bhatti" area of the sanctuary was historically subjected to heavy, unscientific sandstone, badarpur sand, and gravel mining, which left behind deep, abandoned mine pits that have now

transformed into artificial lakes. Restoration efforts focus on rewilding these degraded lands.

Q 125. Ans. B

Explanation:

- **During the 'Makhana Mahotsav 2025' held at Gyan Bhavan in Patna, Union Agriculture and Farmers' Welfare Minister Shivraj Singh Chouhan released a comprehensive report titled "Makhana: From Culture to Prosperity"** (also referred to in Hindi as Makhana: Sanskriti se Samriddhi).
- The report focuses on strategies to boost the rural economy by scaling up the production, processing, branding, and export of makhana (fox nut). It details how makhana—deeply rooted in the cultural and agricultural fabric of the Mithila region—can be transformed into a globally exported 'superfood' to bring significant economic prosperity to the farmers of Bihar.
- **Additional Facts:**
  - ✓ **A Booming Industry:** According to statements made at the Mahotsav, makhana cultivation in Bihar has expanded massively, growing from just 3,000 hectares previously to around 35,000–40,000 hectares today.
  - ✓ Alongside the report, the Minister announced the establishment of a dedicated National Makhana Board and sanctioned ₹475 crore for modernising machinery, developing high-yield varieties (like the 'Sabour Makhana-1'), and establishing processing facilities.
  - ✓ The overarching goal of the initiative, as highlighted in the report, is to transition India from just being the world's largest makhana producer (accounting for over 85–90% of global output) to a dominant exporter of value-added makhana products to markets in Europe, the US, and East Asia.

Q 126. Ans. D

Explanation:

- The **Kessler Syndrome** (also known as collisional cascading) is a scenario proposed by NASA scientist Donald J. Kessler in 1978. It describes a situation where the density of objects in Low Earth Orbit (LEO) becomes so high that a single collision between two objects creates a massive cloud of shrapnel. This debris then strikes other satellites or debris, setting off an unstoppable chain reaction of collisions.
- If this theoretical threshold is crossed, the resulting belt of high-speed space junk could render specific orbital ranges entirely unusable for new satellites, manned spaceflight, or deep-space exploration for generations.
- **Additional Facts:**
  - ✓ **Real-World Precedents:** Events like the 2009 accidental collision between an active US Iridium communication satellite and a defunct Russian Kosmos satellite, along with various kinetic Anti-Satellite (ASAT) weapon tests conducted by different nations, have added thousands of pieces of trackable debris to LEO, increasing the risk of this cascading effect.
  - ✓ **Tracking and Awareness:** To mitigate this risk, space agencies rely on advanced tracking networks. India's ISRO established **Project NETRA** (Network for Space Object Tracking and Analysis), which acts as an early warning system to detect debris and protect Indian satellites from potential collisions.
  - ✓ **Active Debris Removal (ADR):** Space agencies and private companies worldwide are developing technologies to actively clean up orbit. Proposed methods include capturing defunct satellites using robotic arms, nets, harpoons, or magnetic grappling, as well as international guidelines requiring satellites to de-orbit themselves safely at the end of their operational lifespan.

Q 127. Ans. C

Explanation:

- **India's top trading partner (FY 2026) – China (Correctly matched):** According to the latest data from the Commerce Ministry, China overtook the United States to emerge as India's largest trading partner in the financial year 2025-26, with bilateral trade reaching $151.1 billion. The US had previously held the top spot for four consecutive years.
- **India's third-largest oil supplier – Venezuela (Correctly matched):** Amid geopolitical shifts and disruptions surrounding the Strait of Hormuz, Venezuela emerged as India's third-largest crude oil supplier in May 2026, overtaking both Saudi Arabia and the United States due to the cost advantages of its heavy crude. Only Russia and the UAE supplied more oil to India.
- **World's largest recipient of remittances – India (Correctly matched):** According to the Economic Survey 2025-26 and World Bank reports, India consistently remains the world's largest recipient of

remittances, with inflows reaching a record $135.4 billion in FY 2026 (largely driven by an increase in remittances from advanced economies like the US, UK, and UAE).

- **Country that surpassed India as the 5th-largest stock market – Japan (Incorrectly matched):** Japan currently holds the position of the world's **3rd-largest** stock market (behind the US and China). The countries that recently contested the 5th-largest spot and temporarily surpassed India in mid-2026 were **Taiwan** and **South Korea** (driven by the global AI-chip boom), though India later reclaimed the 5th spot.

Q 128. Ans. B

Explanation:

- **Pair 1 (Incorrect):** Telangana is **not** the first state to declare cancer a notifiable disease. Historically, **Tripura** became the first state in India to make cancer a notifiable disease back in 2008. More recently, in January 2026, **Andhra Pradesh** made national headlines by issuing a comprehensive notification and launching a State Cancer Atlas. Telangana only followed suit later in April 2026, becoming one of over 15 states to have implemented this mandate.
- **Pair 2 (Incorrect):** While Paradip Port did briefly hold the top spot in the earlier FY 2023–24 period, it did not secure first place in **FY 2025–26 (FY 2026)**. According to the latest Ministry of Ports, Shipping and Waterways data, the **Deendayal Port Authority** (Gujarat) reclaimed the number one spot by handling 160.11 MT of cargo, closely followed by Paradip Port at 156.45 MT.
- **Pair 3 (Correct):** In April 2026, Adani's **Mundra Port** officially emerged as India's largest automobile export hub. It set a new national record by exporting 6,008 cars in a single vessel through its advanced Roll-on/Roll-off (RoRo) terminal, pushing past traditional automotive export hubs like Chennai and Ennore.
- **Pair 4 (Correct):** Rajasthan's first semiconductor manufacturing cluster was recently inaugurated in **Bhiwadi** (specifically in the Salarpur-Khushkhera area). It is being described as India's first SME-led semiconductor ATMP (Assembly, Testing, Marking, and Packaging) facility, marking the state's formal entry into the high-tech semiconductor ecosystem.

Q 129. Ans. B

Explanation:

- **25th FIFA Football World Cup (2034):** Saudi Arabia was officially appointed by the FIFA Congress as the sole host for the 2034 Men's World Cup.
- **FIFA Women's World Cup (2027):** Brazil was chosen to host the 2027 edition, making it the very first Women's World Cup to be held in South America.
- **FIFA Women's World Cup (2031):** The 2031 tournament will not be hosted by Argentina. It is a four-nation joint hosting effort shared by the **United States, Mexico, Costa Rica, and Jamaica.**
- **FIFA Women's World Cup (2035):** The tournament was awarded jointly to the four separate "home nations", **England, Scotland, Wales, and Northern Ireland**, which all operate as independent football associations and compete as separate national teams.

Q 130. Ans. C

Explanation:

- **The Food and Agriculture Organisation (FAO) of the United Nations designated the official theme for World Pulses Day 2026 (observed on February 10) as "Pulses of the world: from modesty to excellence".**
- The theme highlights the transformative journey of pulses—from being historically viewed as humble, simple, or traditional staples (often associated with low-cost diets) to achieving global recognition for their nutritional excellence, culinary versatility, and vital role in sustainable, climate-resilient agriculture.
- **Additional Facts:**
  - ✓ **Establishment:** Building on the massive success of the FAO-led International Year of Pulses in 2016, the UN General Assembly formally designated February 10 as World Pulses Day starting in 2019.
  - ✓ **The 2026 Host:** The main global event for World Pulses Day 2026 was hosted in collaboration with the Kingdom of Spain in Valladolid, focusing on science, food culture, and sustainable agrifood systems.
  - ✓ **Environmental & Nutritional Value:** Pulses (such as lentils, chickpeas, and beans) naturally fix atmospheric nitrogen into the soil. This unique property significantly reduces the need for synthetic fertilisers, lowering greenhouse gas emissions. Furthermore, their low water footprint makes them a highly climate-smart crop.
  - ✓ **India's Position:** India remains the world's largest producer, consumer, and importer of pulses, heavily relying on them as a primary source of plant-based protein to combat malnutrition.

Q 131. Ans. C

Explanation:

- The **Prakriti Summit 2026** was held from March 19 to 22, 2026, in **New Delhi**.
- PRAKRITI stands for **Promoting Resilience, Awareness, Knowledge, and Resources for Integrating Transformative Initiatives**. The summit served as a prominent national platform for stakeholders to address climate challenges, sustainable development, and India's green transition.
- **Additional Facts:**
  - ✓ **Indian Carbon Market Portal:** A major highlight of the summit was the official launch of the Indian Carbon Market Portal by the government.
  - ✓ **Key Agencies:** The portal was developed by the **Bureau of Energy Efficiency (BEE)** (under the Ministry of Power) in collaboration with the Ministry of Environment, Forest and Climate Change.
  - ✓ **Core Objective:** The summit and the new portal aim to streamline carbon credit trading, boost transparent climate finance, and keep India on track to achieve its ambitious 2035 Nationally Determined Contributions (NDCs) and long-term net-zero emission goals.

Q 132. Ans. D

Explanation:

- Professor **Anantha P. Chandrakasan** became the first Indian-American to be appointed as the Provost of the Massachusetts Institute of Technology (MIT). His appointment was announced in June 2025, and he officially assumed the office on July 1, 2025, succeeding Cynthia Barnhart.
- As Provost, he serves as MIT's chief academic and budget officer. Born in Chennai, India, Chandrakasan is an internationally recognized authority on low-power electronics and energy-efficient circuits. Prior to stepping into the role of Provost, he held several key leadership positions at MIT, including Dean of the School of Engineering (from 2017 to 2025) and MIT's inaugural Chief Innovation and Strategy Officer.

Q 133. Ans. D

Explanation:

- **The 2025 Sakharov Prize for Freedom of Thought was jointly awarded to two imprisoned journalists fighting for democracy and freedom of expression in their respective countries:**
  - ✓ **Andrzej Poczobut (Belarus):** A prominent journalist, blogger, and activist representing the Polish minority in Belarus, who was sentenced to eight years in a penal colony for his outspoken criticism of the Lukashenko regime.
  - ✓ **Mzia Amaglobeli (Georgia):** A leading Georgian journalist and founder of independent media outlets (Batumelebi and Netgazeti), who became the country's first female political prisoner since its independence after being arrested for her involvement in pro-democracy protests.

Q 134. Ans. B

Explanation:

- **Operation True Promise 3** was a large-scale military operation launched by **Iran** (specifically the Islamic Revolutionary Guard Corps) against Israel in June 2025.
  - ✓ The operation was launched in direct retaliation for Israel's preemptive airstrikes (codenamed "Rising Lion"), which targeted Iranian military and nuclear facilities, resulting in the deaths of senior military leaders and scientists.
  - ✓ During Operation True Promise 3, Iran launched massive waves of advanced ballistic missiles and suicide drones aimed at overwhelming Israeli air defence systems, such as the Iron Dome. The strikes targeted key cities, military installations, and intelligence headquarters across Israel.

Q 135. Ans. B

Explanation:

- **Drools** (Drools Pet Food Private Limited) became India's first pet food unicorn in May 2025 after achieving a valuation of over $1 billion.
- The unicorn milestone was reached following a strategic minority stake investment from the Swiss consumer goods giant **Nestlé S.A.** (the parent company of Nestlé India).
- This investment reflects the rapid growth of India's pet care ecosystem, which has been largely driven by a rising demographic of millennial and Gen Z pet parents seeking premium, science-backed nutrition for their pets.
- Founded in 2010 by Fahim Sultan and initially backed by private equity firm L Catterton, Drools currently controls a massive portion of the Indian pet food market by volume (estimated at around 38%) and exports to over 22 countries.

Q 136. Ans. A

Explanation:

- **WWF AI-based conservation app (N. Alim Yusuf):** Kerala-based botanist N. Alim Yusuf developed an AI-powered mobile application (NeophyteID) capable of identifying nearly 100 invasive plant species across the biodiversity-rich Western Ghats. This critical ecological tool earned him the World Wide Fund for Nature (WWF) National Award in 2026.
- **PM-AJAY portal and app (Ministry of Social Justice - for Scheduled Castes):** The Pradhan Mantri Anusuchit Jaati Abhyuday Yojana (PM-AJAY) is an integrated scheme aimed specifically at the socio-economic development of Scheduled Castes (SCs). Its dedicated portal and app enable real-time monitoring, milestone-linked fund flows, and transparent governance for SC welfare.
- **SMILE-Beggary Survey app (Ministry of Social Justice and Empowerment):** Developed under the SMILE scheme (Support for Marginalised Individuals for Livelihood and Enterprise), this app allows district authorities to map and capture real-time data digitally to assist in the comprehensive rehabilitation of persons engaged in begging.
- **Prashast 2.0 app (NCERT):** PRASHAST (Pre-assessment Holistic Screening Tool) was developed by the National Council of Educational Research and Training (NCERT). It is a school-based checklist application that helps teachers identify and refer children with special needs or early-stage disabilities (as recognised under the RPwD Act) for timely support

Q 137. Ans. C

Explanation:

- India's first end-to-end silicon carbide (SiC) semiconductor manufacturing facility has recently been inaugurated in **Bhubaneswar, Odisha**.
- The ₹2,067 crore project is spearheaded by **SiCSem Private Limited** (a subsidiary of Archean Chemical Industries Limited) and was approved under the India Semiconductor Mission (ISM).
- Operating as a fully integrated Assembly, Testing, Marking, and Packaging (ATMP) facility, it marks a watershed moment for India's high-tech ecosystem. The plant will have the capacity to process 60,000 SiC wafers annually, producing high-performance chips crucial for electric vehicles (EVs), renewable energy systems, smart grids, and industrial automation.
- To foster a robust local ecosystem, SiCSem has also partnered with IIT Bhubaneswar to establish a dedicated SiC research and innovation centre to support talent development and continuous technology innovation.

Q 138. Ans. B

Explanation:

- The **United Nations Educational, Scientific and Cultural Organisation (UNESCO)** recently adopted the world's first global ethical framework for neurotechnology. The framework, officially titled the "Recommendation on the Ethics of Neurotechnology," was adopted in November 2025 during its 43rd General Conference.
- **Objective:** As advancements in AI-decoded brain data and consumer neurotech (like brain-reading earbuds or sleep trackers) rapidly accelerate, the framework seeks to establish comprehensive global safeguards to protect human dignity, mental privacy, and cognitive liberty.
- **Key Provisions:** It officially defines "neural data" as a new, highly sensitive category of information. It calls for strict prohibitions on the manipulative use of neural data for commercial, political, or surveillance purposes (such as monitoring workplace productivity without consent) and emphasises the protection of vulnerable groups, particularly children.
- **Significance:** While it is a non-binding legal instrument, it acts as a foundational roadmap for the 194 member states to translate ethical principles into their national regulations and legislation, ensuring neurotechnology improves human well-being without compromising fundamental human rights.

Q 139. Ans. B

Explanation:

- **Statement 1 is incorrect:** The 27th edition of the "Women and Men in India 2025" report was released by the Ministry of Statistics and Programme Implementation (MoSPI), Government of India, not the Ministry of Social Justice and Empowerment.
- **Statement 2 is correct:** The all-India sex ratio at birth has increased from 904 in 2017-19 to 917 in 2021-23, which indicates an improved survival rate for females.
- **Statement 3 is correct:** Between 2021-22 and 2022-23, the Gross Enrolment Ratio at Higher Education improved from 28.5 to 30.2 for females, and from 28.3 to 28.9 for males.

- **Statement 4 is correct:** Women engaged in managerial positions registered a 102.54 percent increase between 2017 and 2025. During the same time period, men engaged in managerial positions saw a lower increase of 73.80 percent.
- **Additional Facts:**
  - ✓ **Infant Mortality:** Between 2008 and 2023, the infant mortality rate recorded a pronounced and sustained decline for both male and female infants.
  - ✓ **Educational Parity:** Gender parity has been successfully achieved across all levels of school education, from the primary level to the higher secondary level.
  - ✓ **Labour Force Participation:** The Labour Force Participation Rate (LFPR) for ages 15 and above has increased for both genders. Rural females experienced the highest increase in LFPR, rising from 37.5 percent in 2022 to 45.9 percent in 2025.
  - ✓ **Summit Context:** The report was officially released on April 29, 2026, during the National Deliberative Summit on "Data for Development" held in Bhubaneswar, Odisha.
  - ✓ **Data Domains:** The report compiles indicators across key domains such as population, health, education, economic participation, decision-making, and violence against women to help inform gender-responsive policies.

Q 140. Ans. C

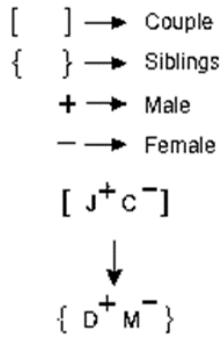
Explanation:

- The **Samriddh Gram Initiative** is a major rural digital transformation project led by the Department of Telecommunications (DoT) under the **Ministry of Communications**.
- **Objective:** Built on the BharatNet high-speed broadband network, the initiative establishes 'Samriddhi Kendras' at the village level. These act as one-stop "phygital" (physical + digital) hubs that deliver integrated services such as telemedicine, smart classrooms, IoT-enabled agriculture, digital banking, and e-governance directly to rural communities.
- The initiative achieved global recognition by winning the WSIS Prize 2026 (awarded by the International Telecommunication Union in Geneva) under the 'Enabling Environment' category for its innovative approach to bridging the rural digital divide.

Q 141. Ans. D

Explanation:

After decoding the above information, we get the following family tree:



Clearly we get C is the mother D.
Hence, option D is correct.

Q 142. Ans. C

Explanation:

- **Logic :** Except for option C, in rest all the sum of these numbers gives a perfect cubic number.
- (857, 871) → 857 + 871 = 1728 = 123
- (1124, 1073) → 1124 + 1073 = 2197 = 133
- (1817, 1558) → 1817 + 1558 = 3375 = 153
- (1396, 1318) → 1396 + 1318 = 2714

Hence, option C is correct.

Q 143. Ans. C

Explanation:

- **On checking option C, we get the meaning word:** CLEVER
- Hence, option C is correct.

Q 144. Ans. A

Explanation:

- 75 * 24 * 72 * 30 * 12 * 7
- **By using option A:**
- 75 × 24 ÷ 72 = 30 − 12 + 7
- 75 ÷ 3 = 25
- 25 = 25
- LHS = RHS
- Hence, option A is correct.

Q 145. Ans. C

Explanation:

- **Logic:** Each consonant of the word is coded as its preceding letter and each vowel is coded as its succeeding letters.
- FOIL → EPJK

- MILD → LKJC
- Similarly, STAKE →RSBJF
- Hence, option C is correct.

Q 146. Ans. A

Explanation:

- According to the question,
- **Let us write all the ratios as fractions:**
- 5 : 8 = $\frac{5}{8}$ = 0.625
- 7 : 10 = $\frac{7}{10}$ = 0.7
- 9 : 14 = $\frac{9}{14}$ = 0.64
- 11 : 16 = $\frac{11}{16}$ = 0.68
- Upon comparison, 0.625 < 0.6429 < 0.6875 < 0.7
- **Therefore, the smallest ratio is 5 :** 8.

Q 147. Ans. C

Explanation:

- The total cost price of both items is the same; therefore,
- Total loss = 25% - 15% = 10%
- Loss (in rupees) = 1200 × 10% = ₹120

Q 148. Ans. A

Explanation:

- According to the question,
- **The three numbers are in the ratio 3 :** 2 : 5.
- Let the numbers be 3x, 2x, and 5x.
- Given that HCF = 12
- Therefore, the numbers are 36, 24, and 60.
- Now,
- Largest number = 60
- Second smallest number = 36
- Therefore, the difference = 60 − 36 = 24

Q 149. Ans. A

Explanation:

- According to the question,
- **Rahul :** Seema = 125 : 100 = 5 : 4
- **Mohan :** Amit = 1 : 2
- Rahul's salary = ₹10,000
- Seema's salary = 10,000 × 4/5 = ₹8,000
- Amit's salary is ₹3,000 less than Seema's salary.
- Amit's salary = 8,000 − 3,000 = ₹5,000
- Mohan's salary = 1/2 × 5,000 = ₹2,500

Q 150. Ans. C

Explanation:

According to the question,

- Time taken to cover a distance of 108 km = 90 minutes = 3/2 hours
- Speed of the bike = 108 ÷ (3/2) = 108 × 2/3 = 72 km/hr
- New speed = 72 - 24 = 48 km/hr
- Required time = 108/48 = 9/4 hours
- = 9/4 × 60 minutes
- = 135 minutes